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Payload retrieval apparatus with internal unlocking feature and security features for use with a UAV

Abstract

A payload retrieval apparatus having a base, an autoloader assembly mounted to the base including: a payload holder configured to hold a payload for retrieval by a UAV, a channel coupled to the payload holder configured to direct a payload retriever suspended from the UAV to the payload holder, and a locking feature configured to lock access to the payload on the payload holder that includes has a movable end that extends through a wall of the channel into an interior of the channel, wherein when the payload retriever contacts the movable end of the locking member, the movable end moves outwardly thereby unlocking the payload on the payload holder, wherein the payload holder is positioned such that when the payload retriever exits the channel, the payload retriever engages a handle of the payload and removes the payload from the payload holder.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3270895	12/1965	Stewart	D12/323	B64B 1/00
3807577	12/1973	Bell	414/23	B66C 21/00
3865251	12/1974	Langford	244/33	B66C 21/00
4055316	12/1976	Chipper	104/23.1	B64B 1/50
5080302	12/1991	Hoke	244/33	B64B 1/50
6091161	12/1999	Dehlsen	290/43	F03B 13/10
7183663	12/2006	Roberts	290/55	F03D 9/25
7287724	12/2006	Hase	244/33	B64B 1/50
8205835	12/2011	Wiley	244/33	B63B 35/44
8602349	12/2012	Petrov	244/17.23	G05D 1/0866
8907516	12/2013	Carroll	290/55	F03D 5/00
9043052	12/2014	So	701/3	G08G 9/00
9045218	12/2014	Childress	N/A	B60L 9/00
9174733	12/2014	Burgess	N/A	B64D 1/12
9561852	12/2016	Beaman	N/A	B64C 37/02
10577105	12/2019	Prager	N/A	B64D 1/22
2011/0180667	12/2010	O'Brien	244/135R	B64U 10/60
2013/0233964	12/2012	Woodworth	244/175	B64U 10/60
2013/0306791	12/2012	McGeer	244/110F	B64F 1/02
2016/0059963	12/2015	Burgess	N/A	N/A
2017/0073071	12/2016	Salzmann	N/A	B64U 10/10
2017/0121023	12/2016	High	N/A	B64D 9/00
2018/0141682	12/2017	Blake	N/A	N/A
2019/0084674	12/2018	Jones	N/A	N/A
2019/0100314	12/2018	Prager	N/A	N/A
2019/0185162	12/2018	Prager	N/A	N/A
2019/0256210	12/2018	Prager	N/A	N/A
2019/0375504	12/2018	Schmalzried	N/A	N/A
2022/0073214	12/2021	Liske	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC

115489735 12/2021 CN N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion for Int. App. No. PCT/US2023/086400, mailed Nov. 8, 2024. cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Patent Application No. 63/477,869, filed Dec. 30, 2022, the contents of which are incorporated by reference.

BACKGROUND
(1) An uncrewed vehicle, which may also be referred to as an autonomous vehicle, is a vehicle capable of travel without a physically-present human operator. An uncrewed vehicle may operate in a remote-control mode, in an autonomous mode, or in a partially autonomous mode.
(2) An uncrewed aerial vehicle (UAV) may be used to deliver a payload to, or retrieve a payload from, an individual or business. The term “uncrewed aerial vehicle” is synonymous and interchangeable with the term “unmanned aerial vehicle” which has often been used. In some operations, once the UAV arrives at a retrieval site, the UAV may land or remain in a hover position. At this point, a person at the retrieval site may secure the payload to the UAV at an end of a tether attached to a winch mechanism positioned with the UAV, or to the UAV itself. For example, the payload may have a handle that may be secured to a device at the end of the winch, or a handle that may be secured within the UAV. However, this scenario has a number of drawbacks. In particular, if the UAV is late for arrival at the retrieval site, the person designated for securing the payload to be retrieved by the UAV may have to wait a period of time before the UAV arrives, resulting in undesirable waiting time. Similarly, if the UAV arrives and the person designated to secure the payload to be retrieved to the UAV is delayed or fails to show up, the UAV may have to wait in a hover mode or on the ground until the designated person arrives to secure the payload to the UAV, resulting in undesirable delay and expenditure of energy by the UAV as the UAV waits for the designated person to arrive, and also resulting in undesirable delay in the subsequent delivery of the payload at a delivery site.
(3) As a result, it would be desirable to provide for the automated pickup of a payload by the UAV, where the UAV may automatically pick up the payload without the need for a designated person to secure the payload to the UAV at the retrieval site. Such automated pickup of the payload by the UAV would advantageously eliminate the need for a designated person to secure the payload to the UAV and eliminate potential delays associated with the late arrival of the UAV or designated person at the retrieval site.
(4) In addition, it would be desirable to provide for increased security for a payload and its contents positioned on a payload retrieval apparatus as the payload awaits retrieval by a UAV.
SUMMARY
(5) The present embodiments are directed to a payload retrieval apparatus. A payload is positioned on a payload holder in an autoloader assembly ready for retrieval by a UAV. A locking feature is used to lock the payload and its contents to prevent theft or tampering with the payload or its contents as the payload awaits retrieval by a UAV. In some embodiments, an unlocking feature is

included wherein a payload retriever traveling through a channel of the payload retrieval apparatus releases the locking feature to allow for retrieval of the payload by the payload retriever when the payload retriever exits the channel

(6) In one aspect, a payload retrieval apparatus is provided having a base; an autoloader assembly mounted to the base including: a payload holder configured to hold a payload for retrieval by an uncrewed aerial vehicle (UAV), a channel coupled to the payload holder and configured to direct a payload retriever suspended from the UAV to the payload holder, the channel having a tether slot adapted for receiving a tether having a first end attached to the UAV and a second end attached to the payload retriever, wherein the channel is adapted to receive the payload retriever, and the tether slot of the channel allows for passage of the tether during movement of the payload retriever through the channel; and a locking feature configured to lock access to the payload on the payload holder, the locking feature including a locking member that has a movable end, wherein the locking member has a first position in which the movable end extends through a wall of the channel into an interior of the channel for engagement by the payload retriever as it moves through the channel, wherein when the payload retriever contacts the movable end of the locking member, the movable end moves outwardly thereby moving the locking member to a second position and unlocking the payload on the payload holder; wherein the payload holder is positioned such that when the payload retriever exits the channel, the payload retriever engages a handle of the payload and removes the payload from the payload holder.

(7) In another aspect, a method is provided including (i) providing a payload retrieval apparatus having a base; an autoloader assembly mounted to the base including: a payload holder configured to hold a payload for retrieval by an uncrewed aerial vehicle (UAV), a channel coupled to the payload holder and configured to direct a payload retriever suspended from the UAV to the payload holder, the channel having a tether slot adapted for receiving a tether having a first end attached to the UAV and a second end attached to the payload retriever, wherein the channel is adapted to receive the payload retriever, and the tether slot of the channel allows for passage of the tether during movement of the payload retriever through the channel; and a locking feature configured to lock access to the payload on the payload holder, the locking feature including a locking member that has a movable end, wherein the locking member has a first position in which the movable end extends through a wall of the channel into an interior of the channel for engagement by the payload retriever as it moves through the channel, wherein when the payload retriever contacts the movable end of the locking member, the movable end moves outwardly thereby moving the locking member to a second position and unlocking the payload on the payload holder; and wherein the payload holder is positioned such that when the payload retriever exits the channel, the payload retriever engages a handle of the payload and removes the payload from the payload holder; and (ii) causing the payload to pass through the channel and engage the movable end of the locking member to unlock access to the payload holder and payload.

(8) In yet a further aspect, a payload retrieval apparatus is provided having a base; an autoloader assembly mounted to the base including: a payload holder configured to hold a payload for retrieval by an uncrewed aerial vehicle (UAV), a channel coupled to the payload holder and configured to direct a payload retriever suspended from the UAV to the payload holder, the channel having a tether slot adapted for receiving a tether having a first end attached to the UAV and a second end attached to the payload retriever, wherein the channel is adapted to receive the payload retriever, and the tether slot of the channel allows for passage of the tether during movement of the payload retriever through the channel; and a locking feature configured to lock access to the payload on the payload holder; and wherein the payload holder is positioned such that when the payload retriever exits the channel, the payload retriever engages a handle of the payload and removes the payload from the payload holder.

(9) In another aspect, a method is provided including (i) providing a base; an autoloader assembly mounted to the base including: a payload holder configured to hold a payload for retrieval by an

uncrewed aerial vehicle (UAV), a channel coupled to the payload holder and configured to direct a payload retriever suspended from the UAV to the payload holder, the channel having a tether slot adapted for receiving a tether having a first end attached to the UAV and a second end attached to the payload retriever, wherein the channel is adapted to receive the payload retriever, and the tether slot of the channel allows for passage of the tether during movement of the payload retriever through the channel; and a locking feature configured to lock access to the payload on the payload holder; and wherein the payload holder is positioned such that when the payload retriever exits the channel, the payload retriever engages a handle of the payload and removes the payload from the payload holder; and (ii) unlocking the locking feature to provide access to the payload holder and payload for payload retrieval by the payload retriever.

(10) In another aspect, a payload retrieval apparatus is provided including a base; an autoloader assembly mounted to the base including: a payload holder configured to hold a payload for retrieval by an uncrewed aerial vehicle (UAV), a channel coupled to the payload holder and configured to direct a payload retriever suspended from the UAV to the payload holder, the channel having a tether slot adapted for receiving a tether having a first end attached to the UAV and a second end attached to the payload retriever, wherein the channel is adapted to receive the payload retriever, and the tether slot of the channel allows for passage of the tether during movement of the payload retriever through the channel; and a locking feature configured to lock access to the payload on the payload holder, the locking feature including a switch disposed along the channel and configured to be activated by the payload retriever as it moves through the channel, wherein when the payload retriever activates the switch, the locking member unlocks the payload on the payload holder; and wherein the payload holder is positioned such that when the payload retriever exits the channel, the payload retriever engages a handle of the payload and removes the payload from the payload holder.

(11) These as well as other aspects, advantages, and alternatives will become apparent to those of ordinary skill in the art by reading the following detailed description with reference where appropriate to the accompanying drawings. Further, it should be understood that the description provided in this summary section and elsewhere in this document is intended to illustrate the claimed subject matter by way of example and not by way of limitation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1**A is a simplified illustration of an unmanned aerial vehicle **100**, according to an example embodiment.

(2) FIG. **1**B is a simplified illustration of an unmanned aerial vehicle **120**, according to an example embodiment.

(3) FIG. **1**C is a simplified illustration of an unmanned aerial vehicle **140**, according to an example embodiment.

(4) FIG. **1**D is a simplified illustration of an unmanned aerial vehicle **160**, according to an example embodiment.

(5) FIG. **1**E is a simplified illustration of an unmanned aerial vehicle **180**, according to an example embodiment.

(6) FIG. **2** is a simplified block diagram illustrating components of an unmanned aerial vehicle **200**, according to an example embodiment.

(7) FIG. **3** is a simplified block diagram illustrating a UAV system, according to an example embodiment.

(8) FIG. **4**A shows a side view of a payload delivery apparatus with a payload secured to a UAV, according to an example embodiment.

(9) FIG. **4**B shows a side view of the payload delivery apparatus shown in FIG. **4**A lowering the payload to a delivery location.
(10) FIG. **4**C shows a side view of the payload delivery apparatus shown in FIGS. **4**A and **4**B after delivering the payload to the delivery location.
(11) FIG. **5**A shows a perspective view of a payload delivery apparatus **500** including payload **510**, according to an example embodiment.
(12) FIG. **5**B is a cross-sectional side view of payload delivery apparatus **500** and payload **510** shown in FIG. **5**A.
(13) FIG. **5**C is a side view of payload delivery apparatus **500** and payload **510** shown in FIGS. **5**A and **5**B.
(14) FIG. **6**A is a perspective view of payload coupling apparatus **800**, according to an example embodiment.
(15) FIG. **6**B is a side view of payload coupling apparatus **800** shown in FIG. **6**A.
(16) FIG. **6**C is a front view of payload coupling apparatus **800** shown in FIGS. **6**A and **6**B.
(17) FIG. **7** is a perspective view of payload coupling apparatus **800** shown in FIGS. **6**A-**6**C, prior to insertion into a payload coupling apparatus receptacle positioned in the fuselage of a UAV.
(18) FIG. **8** is another perspective view of payload coupling apparatus **800** shown in FIGS. **6**A-**6**C, prior to insertion into a payload coupling apparatus receptacle positioned in the fuselage of a UAV.
(19) FIG. **9** shows a perspective view of a recessed restraint slot and payload coupling apparatus receptacle positioned in a fuselage of a UAV.
(20) FIG. **10**A shows a side view of a payload delivery apparatus **500** with a handle **511** of payload **510** secured within a payload coupling apparatus **800** as the payload **510** moves downwardly prior to touching down for delivery.
(21) FIG. **10**B shows a side view of payload delivery apparatus **500** after payload **510** has landed on the ground showing payload coupling apparatus **800** decoupled from handle **511** of payload **510**.
(22) FIG. **10**C shows a side view of payload delivery apparatus **500** with payload coupling apparatus **800** moving away from handle **511** of payload **510**.
(23) FIG. **11**A is a side view of handle **511** of payload **510** having openings **524** and **526** adapted to receive pins positioned on a payload holder, according to an example embodiment.
(24) FIG. **11**B is a side view of handle **511**′ of a payload having magnets **524**′ and **526**′ positioned thereon for magnetic engagement with a payload holder, according to an example embodiment.
(25) FIG. **12** shows a pair of locking pins **570**, **572** extending through holes **524** and **526** in handle **511** of payload **510** to secure the handle **511** and top of payload **510** within the fuselage of a UAV, or to secure the handle **511** to a payload holder on a payload retrieval apparatus.
(26) FIG. **13**A is a side view of payload coupling apparatus **800**′ with a slot **808** positioned above lip **806**′, according to an example embodiment.
(27) FIG. **13**B is a side view of payload coupling apparatus **800**′ after lip **806**′ has been moved outwardly to facilitate engagement with a handle of a payload.
(28) FIG. **13**C is a side view of payload coupling apparatus **800** having a plurality of magnets **830** positioned thereon, according to an example embodiment.
(29) FIG. **13**D is a side view of payload coupling apparatus **900** having a weighted side **840**, according to an example embodiment.
(30) FIG. **14** is a perspective view of payload retrieval apparatus **1000** having a payload **510** positioned thereon, according to an example embodiment.
(31) FIG. **15** is another perspective view of payload retrieval apparatus **1000** and payload **510** shown in FIG. **14**.
(32) FIG. **16** is a further perspective view of payload retrieval apparatus **1000** and payload **510** shown in FIGS. **14** and **15**.
(33) FIG. **17** shows a sequence of steps A-D performed in the retrieval of payload **510** from payload retrieval apparatus **1000** shown in FIGS. **14-16**.

(34) FIG. **18** is a perspective view of payload retrieval apparatus **1000** shown in FIGS. **14-17** with a payload loading apparatus **1080** having a plurality of payloads positioned thereon, according to an example embodiment.
(35) FIG. **19** is a perspective view of channel **1050** of the payload retrieval apparatus **1000** shown in FIGS. **14-16** with a payload retriever **800** positioned therein.
(36) FIG. **20** is a perspective view of channel **1050** of the payload retrieval apparatus **1000** shown in FIGS. **14-16** with a payload retriever **800**″.
(37) FIG. **21**A is a cross-sectional view of channel **1050**, according to an example embodiment.
(38) FIG. **21**B is a side view of channel **1050** having a spring **1059** biased against end **1057** thereof.
(39) FIG. **22** is a side view of payload retrieval apparatus **1400**.
(40) FIG. **23** is a top view of payload retrieval apparatus **1400**.
(41) FIGS. **24**A-E illustrate a sequence of steps used to automatically pick up payload **510**.
(42) FIG. **25**A is a perspective view of payload retrieval apparatus **1480**.
(43) FIG. **25**B is a side view of payload retrieval apparatus **1480**.
(44) FIG. **25**C is a side view of an end of payload retrieval apparatus **1480** with payload **510** positioned on curved portion **1439**.
(45) FIG. **25**D is a perspective view of an end of payload retrieval apparatus **1480** with payload **510** positioned on curved portion **1439**.
(46) FIG. **25**E shows a perspective view of payload retrieval apparatus **1500**.
(47) FIG. **26** is a perspective view of payload retrieval apparatus **1480**.
(48) FIG. **27**A shows perspective views of rotational spring loaded pusher **1600**.
(49) FIG. **27**B shows a side view of leaf spring **1640**.
(50) FIG. **27**C shows a side view of linear spring plunger **1650**.
(51) FIG. **28** is a perspective view of payload retrieval apparatus **1700**.
(52) FIGS. **29**A-B show perspective and side views of spring loaded plunger pin **1484**.
(53) FIGS. **30**A-B show side views of protrusions **1519**.
(54) FIGS. **31**A-B show side and perspective views of curved portion **1439**.
(55) FIGS. **32**A-C show side and perspective views of curved portion **1439**.
(56) FIGS. **33**A-B show perspective views of curved portion **1439**.
(57) FIGS. **34**A-E show various perspective views of pivoting carriage **1700**.
(58) FIG. **35** shows a side view of payload retrieval apparatus **1000**.
(59) FIG. **36** shows a front view of payload retrieval apparatus **1900**.
(60) FIG. **37** shows a perspective side view of payload retrieval apparatus **1900**.
(61) FIG. **38** shows a rear view of payload retrieval apparatus **1900**.
(62) FIG. **39** shows a top view of payload retrieval apparatus **1900**.
(63) FIG. **40**A shows a partial top view of payload retrieval apparatus **1900**.
(64) FIG. **40**B shows a partial rear perspective view of payload retrieval apparatus **1900**.
(65) FIG. **40**C shows a partial rear perspective view of payload retrieval apparatus **1900**.
(66) FIG. **40**D shows a partial rear perspective view of payload retrieval apparatus **1900**.
(67) FIG. **41**A shows a perspective view of payload retrieval apparatus **1950**.
(68) FIG. **41**B shows a front view of payload retrieval apparatus **1950**.
(69) FIG. **41**C shows a partial side view of payload retrieval apparatus **1950**.
(70) FIG. **42** is a close-up rear view of payload retrieval apparatus **1900**.
(71) FIG. **43** is a close-up perspective rear view of payload retrieval apparatus **1900**.
(72) FIG. **44** is a close-up perspective front view of payload retrieval apparatus **1900**.
(73) FIG. **45**A is a perspective view of payload retriever **800** passing through tether channel passage **1962** of channel **1960** towards locking member **2040**.
(74) FIG. **45**B is a perspective view of payload retriever **800** passing through tether channel passage **1964** of channel **1960** and a movable end of locking member **2040** to move locking member **2040** to an unlocked position where locking strap **2055** is moved away from the payload

handle **511** of payload **2150**.
(75) FIG. **45**C is a perspective view of payload retriever **800** exiting the channel passage **1964** of channel **1960** and engaging handle **511** of payload **2150**.
(76) FIG. **45**D is a perspective view of payload retriever **800** after removal of payload **2150** from pins **570** and **572**.
(77) FIG. **46**A is a side view of payload retriever **800** passing through tether channel passage **1433** towards locking member **2039**.
(78) FIG. **46**B is a perspective view of locking strap **2090** positioned over handle **511** and pins **570** and **572**.
(79) FIG. **46**C is another perspective view of locking strap **2090** positioned over handle **511** and pins **570** and **572**.
(80) FIG. **46**D is a perspective view of locking strap **2090** in an open position.
(81) FIG. **47**A is a perspective view of combination payload retrieval and package pickup apparatus **2000** with payload **2150** positioned on payload holder **2060**.
(82) FIG. **47**B is a side view of combination payload retrieval and package pickup apparatus **2000** shown in FIG. **47**A.
(83) FIG. **47**C is a perspective view of combination payload retrieval and package pickup apparatus **2000** after payload holder **2060** has been retracted into extending member **2050**.
(84) FIG. **47**D is a side view of combination payload retrieval and package pickup apparatus **2000** shown in FIG. **47**C.
(85) FIG. **48**A is a perspective view of combination payload retrieval and package pickup apparatus **2000** with payload **2150** positioned on payload holder **2060**.
(86) FIG. **48**B is a side view of combination payload retrieval and package pickup apparatus **2000** shown in FIG. **48**A.
(87) FIG. **48**C is a perspective view of combination payload retrieval and package pickup apparatus **2000** after payload holder **2060** and payload **2150** have been rotated 180 degrees into extending member **2050**.
(88) FIG. **48**D is a side view of combination payload retrieval and package pickup apparatus **2000** shown in FIG. **48**C.
(89) FIG. **49**A is a perspective view of locking gate **2080** positioned above payload **2150**.
(90) FIG. **49**B is a side view of locking gate **2080** positioned above payload **2150**.
(91) FIG. **49**C is a perspective view of locking gate **2080** positioned around payload **2150**.
(92) FIG. **49**D is a side view of locking gate **2080** positioned around payload **2150**.
(93) FIG. **50**A is a perspective view of a portion of a payload retrieval apparatus **5000** in an open position.
(94) FIG. **50**B is a perspective view of the payload retrieval apparatus **5000** in a closed position.
(95) FIG. **51**A is a cross-sectional top view of a mechanism **5100** for operating a latch.
(96) FIG. **51**B is a cross-sectional top view of the mechanism **5100** being fully opened by a payload retriever **800**.
(97) FIG. **51**C is a cross-sectional top view of the mechanism **5100** being only partially operated by an object **5111**.
(98) FIG. **52**A is a cross-sectional top view of a mechanism **5200** for opening a latch.
(99) FIG. **52**B is a cross-sectional top view of the mechanism **5200** being fully opened by a payload retriever **800**.
(100) FIG. **53** is a perspective view of a portion of a payload retrieval apparatus **5300**.
(101) FIG. **54**A is a cross-sectional side view of a locking bracket **5460** in a closed position.
(102) FIG. **54**B is a cross-sectional side view of the locking bracket **5460** in an open position.
(103) FIG. **55**A is a cross-sectional side view of a locking bracket **5560** in an open position.
(104) FIG. **55**B is a cross-sectional side view of the locking bracket **5560** in a closed position.
(105) FIG. **56**A is an exploded perspective view of a locking structure **5600** on a payload retrieval

apparatus.
(106) FIG. **56**B is a perspective view of locking structure **5600** in a closed position.
(107) FIG. **56**C is a cross-sectional side view of locking structure **5600** in a closed position.
(108) FIG. **57**A is an exploded perspective view of another locking structure **5700** on a payload retrieval apparatus.
(109) FIG. **57**B shows top, back, and side views of a rotating component **5788** of locking structure **5700**.

DETAILED DESCRIPTION

(110) Exemplary methods and systems are described herein. It should be understood that the word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation or feature described herein as “exemplary” or “illustrative” is not necessarily to be construed as preferred or advantageous over other implementations or features. In the figures, similar symbols typically identify similar components, unless context dictates otherwise. The example implementations described herein are not meant to be limiting. It will be readily understood that the aspects of the present disclosure, as generally described herein, and illustrated in the figures, can be arranged, substituted, combined, separated, and designed in a wide variety of different configurations, all of which are contemplated herein.
(111) Furthermore, the particular arrangements shown in the Figures should not be viewed as limiting. It should be understood that other embodiments might include more or less of each element shown in a given Figure. Further, some of the illustrated elements may be combined or omitted. Yet further, an example embodiment may include elements that are not illustrated in the Figures.
(112) The present embodiments are related to the use of UAVs that are used to carry a payload to be delivered or retrieved. As examples, UAVs may be used to deliver or retrieve a payload to or from an individual or business. In operation the payload to be delivered is secured to the UAV and the UAV is then flown to the desired delivery site. The payload may be secured beneath the UAV, positioned within the UAV, or positioned partially within the UAV, as the UAV flies to the delivery site. Once the UAV arrives at the delivery site, the UAV may land to deliver the payload, or may be operated in a hover mode while the payload is dropped or lowered from the UAV towards the delivery site using a tether and a winch mechanism positioned within the UAV.
(113) Herein, the terms “uncrewed aerial vehicle” and “UAV” refer to any autonomous or semi-autonomous vehicle that is capable of performing some functions without a physically present human pilot.
(114) A UAV can take various forms. For example, a UAV may take the form of a fixed-wing aircraft, a glider aircraft, a tail-sitter aircraft, a jet aircraft, a ducted fan aircraft, a lighter-than-air dirigible such as a blimp or steerable balloon, a rotorcraft such as a helicopter or multicopter, and/or an ornithopter, among other possibilities. Further, the terms “drone,” “unmanned aerial vehicle system” (UAVS), or “unmanned aerial system” (UAS) may also be used to refer to a UAV.
(115) FIG. **1**A is an isometric view of an example UAV **100**. UAV **100** includes wing **102**, booms **104**, and a fuselage **106**. Wings **102** may be stationary and may generate lift based on the wing shape and the UAV's forward airspeed. For instance, the two wings **102** may have an airfoil-shaped cross section to produce an aerodynamic force on UAV **100**. In some embodiments, wing **102** may carry horizontal propulsion units **108**, and booms **104** may carry vertical propulsion units **110**. In operation, power for the propulsion units may be provided from a battery compartment **112** of fuselage **106**. In some embodiments, fuselage **106** also includes an avionics compartment **114**, an additional battery compartment (not shown) and/or a delivery unit (not shown, e.g., a winch system) for handling the payload. In some embodiments, fuselage **106** is modular, and two or more compartments (e.g., battery compartment **112**, avionics compartment **114**, other payload and delivery compartments) are detachable from each other and securable to each other (e.g., mechanically, magnetically, or otherwise) to contiguously form at least a portion of fuselage **106**.

(116) In some embodiments, booms **104** terminate in rudders **116** for improved yaw control of UAV **100**. Further, wings **102** may terminate in wing tips **117** for improved control of lift of the UAV.
(117) In the illustrated configuration, UAV **100** includes a structural frame. The structural frame may be referred to as a "structural H-frame" or an "H-frame" (not shown) of the UAV. The H-frame may include, within wings **102**, a wing spar (not shown) and, within booms **104**, boom carriers (not shown). In some embodiments the wing spar and the boom carriers may be made of carbon fiber, hard plastic, aluminum, light metal alloys, or other materials. The wing spar and the boom carriers may be connected with clamps. The wing spar may include pre-drilled holes for horizontal propulsion units **108**, and the boom carriers may include pre-drilled holes for vertical propulsion units **110**.
(118) In some embodiments, fuselage **106** may be removably attached to the H-frame (e.g., attached to the wing spar by clamps, configured with grooves, protrusions or other features to mate with corresponding H-frame features, etc.). In other embodiments, fuselage **106** similarly may be removably attached to wings **102**. The removable attachment of fuselage **106** may improve quality and or modularity of UAV **100**. For example, electrical/mechanical components and/or subsystems of fuselage **106** may be tested separately from, and before being attached to, the H-frame. Similarly, printed circuit boards (PCBs) **118** may be tested separately from, and before being attached to, the boom carriers, therefore eliminating defective parts/subassemblies prior to completing the UAV. For example, components of fuselage **106** (e.g., avionics, battery unit, delivery units, an additional battery compartment, etc.) may be electrically tested before fuselage **106** is mounted to the H-frame. Furthermore, the motors and the electronics of PCBs **118** may also be electrically tested before the final assembly. Generally, the identification of the defective parts and subassemblies early in the assembly process lowers the overall cost and lead time of the UAV. Furthermore, different types/models of fuselage **106** may be attached to the H-frame, therefore improving the modularity of the design. Such modularity allows these various parts of UAV **100** to be upgraded without a substantial overhaul to the manufacturing process.
(119) In some embodiments, a wing shell and boom shells may be attached to the H-frame by adhesive elements (e.g., adhesive tape, double-sided adhesive tape, glue, etc.). Therefore, multiple shells may be attached to the H-frame instead of having a monolithic body sprayed onto the H-frame. In some embodiments, the presence of the multiple shells reduces the stresses induced by the coefficient of thermal expansion of the structural frame of the UAV. As a result, the UAV may have better dimensional accuracy and/or improved reliability.
(120) Moreover, in at least some embodiments, the same H-frame may be used with the wing shell and/or boom shells having different size and/or design, therefore improving the modularity and versatility of the UAV designs. The wing shell and/or the boom shells may be made of relatively light polymers (e.g., closed cell foam) covered by the harder, but relatively thin, plastic skins.
(121) The power and/or control signals from fuselage **106** may be routed to PCBs **118** through cables running through fuselage **106**, wings **102**, and booms **104**. In the illustrated embodiment, UAV **100** has four PCBs, but other numbers of PCBs are also possible. For example, UAV **100** may include two PCBs, one per the boom. The PCBs carry electronic components **119** including, for example, power converters, controllers, memory, passive components, etc. In operation, propulsion units **108** and **110** of UAV **100** are electrically connected to the PCBs.
(122) Many variations on the illustrated UAV are possible. For instance, fixed-wing UAVs may include more or fewer rotor units (vertical or horizontal), and/or may utilize a ducted fan or multiple ducted fans for propulsion. Further, UAVs with more wings (e.g., an "x-wing" configuration with four wings), are also possible. Although FIG. **1** illustrates two wings **102**, two booms **104**, two horizontal propulsion units **108**, and six vertical propulsion units **110** per boom **104**, it should be appreciated that other variants of UAV **100** may be implemented with more or less of these components. For example, UAV **100** may include four wings **102**, four booms **104**, and

more or less propulsion units (horizontal or vertical).

(123) Similarly, FIG. **1**B shows another example of a fixed-wing UAV **120**. The fixed-wing UAV **120** includes a fuselage **122**, two wings **124** with an airfoil-shaped cross section to provide lift for the UAV **120**, a vertical stabilizer **126** (or fin) to stabilize the plane's yaw (turn left or right), a horizontal stabilizer **128** (also referred to as an elevator or tailplane) to stabilize pitch (tilt up or down), landing gear **130**, and a propulsion unit **132**, which can include a motor, shaft, and propeller.

(124) FIG. **1**C shows an example of a UAV **140** with a propeller in a pusher configuration. The term “pusher” refers to the fact that a propulsion unit **142** is mounted at the back of the UAV and “pushes” the vehicle forward, in contrast to the propulsion unit being mounted at the front of the UAV. Similar to the description provided for FIGS. **1**A and **1**B, FIG. **1**C depicts common structures used in a pusher plane, including a fuselage **144**, two wings **146**, vertical stabilizers **148**, and the propulsion unit **142**, which can include a motor, shaft, and propeller.

(125) FIG. **1**D shows an example of a tail-sitter UAV **160**. In the illustrated example, the tail-sitter UAV **160** has fixed wings **162** to provide lift and allow the UAV **160** to glide horizontally (e.g., along the x-axis, in a position that is approximately perpendicular to the position shown in FIG. **1**D). However, the fixed wings **162** also allow the tail-sitter UAV **160** to take off and land vertically on its own.

(126) For example, at a launch site, the tail-sitter UAV **160** may be positioned vertically (as shown) with its fins **164** and/or wings **162** resting on the ground and stabilizing the UAV **160** in the vertical position. The tail-sitter UAV **160** may then take off by operating its propellers **166** to generate an upward thrust (e.g., a thrust that is generally along the y-axis). Once at a suitable altitude, the tail-sitter UAV **160** may use its flaps **168** to reorient itself in a horizontal position, such that its fuselage **170** is closer to being aligned with the x-axis than the y-axis. Positioned horizontally, the propellers **166** may provide forward thrust so that the tail-sitter UAV **160** can fly in a similar manner as a typical airplane.

(127) Many variations on the illustrated fixed-wing UAVs are possible. For instance, fixed-wing UAVs may include more or fewer propellers, and/or may utilize a ducted fan or multiple ducted fans for propulsion. Further, UAVs with more wings (e.g., an “x-wing” configuration with four wings), with fewer wings, or even with no wings, are also possible.

(128) As noted above, some embodiments may involve other types of UAVs, in addition to or in the alternative to fixed-wing UAVs. For instance, FIG. **1**E shows an example of a rotorcraft that is commonly referred to as a multicopter **180**. The multicopter **180** may also be referred to as a quadcopter, as it includes four rotors **182**. It should be understood that example embodiments may involve a rotorcraft with more or fewer rotors than the multicopter **180**. For example, a helicopter typically has two rotors. Other examples with three or more rotors are possible as well. Herein, the term “multicopter” refers to any rotorcraft having more than two rotors, and the term “helicopter” refers to rotorcraft having two rotors.

(129) Referring to the multicopter **180** in greater detail, the four rotors **182** provide propulsion and maneuverability for the multicopter **180**. More specifically, each rotor **182** includes blades that are attached to a motor **184**. Configured as such, the rotors **182** may allow the multicopter **180** to take off and land vertically, to maneuver in any direction, and/or to hover. Further, the pitch of the blades may be adjusted as a group and/or differentially, and may allow the multicopter **180** to control its pitch, roll, yaw, and/or altitude.

(130) It should be understood that references herein to an “uncrewed” aerial vehicle or UAV can apply equally to autonomous and semi-autonomous aerial vehicles. In an autonomous implementation, all functionality of the aerial vehicle is automated; e.g., pre-programmed or controlled via real-time computer functionality that responds to input from various sensors and/or pre-determined information. In a semi-autonomous implementation, some functions of an aerial vehicle may be controlled by a human operator, while other functions are carried out

autonomously. Further, in some embodiments, a UAV may be configured to allow a remote operator to take over functions that can otherwise be controlled autonomously by the UAV. Yet further, a given type of function may be controlled remotely at one level of abstraction and performed autonomously at another level of abstraction. For example, a remote operator could control high level navigation decisions for a UAV, such as by specifying that the UAV should travel from one location to another (e.g., from a warehouse in a suburban area to a delivery address in a nearby city), while the UAV's navigation system autonomously controls more fine-grained navigation decisions, such as the specific route to take between the two locations, specific flight controls to achieve the route and avoid obstacles while navigating the route, and so on.

(131) More generally, it should be understood that the example UAVs described herein are not intended to be limiting. Example embodiments may relate to, be implemented within, or take the form of any type of unmanned aerial vehicle.

(132) FIG. **2** is a simplified block diagram illustrating components of a UAV **200**, according to an example embodiment. UAV **200** may take the form of, or be similar in form to, one of the UAVs **100**, **120**, **140**, **160**, and **180** described in reference to FIGS. **1**A-**1**E. However, UAV **200** may also take other forms.

(133) UAV **200** may include various types of sensors, and may include a computing system configured to provide the functionality described herein. In the illustrated embodiment, the sensors of UAV **200** include an inertial measurement unit (IMU) **202**, ultrasonic sensor(s) **204**, and a GPS **206**, among other possible sensors and sensing systems.

(134) In the illustrated embodiment, UAV **200** also includes one or more processors **208**. A processor **208** may be a general-purpose processor or a special purpose processor (e.g., digital signal processors, application specific integrated circuits, etc.). The one or more processors **208** can be configured to execute computer-readable program instructions **212** that are stored in the data storage **210** and are executable to provide the functionality of a UAV described herein.

(135) The data storage **210** may include or take the form of one or more computer-readable storage media that can be read or accessed by at least one processor **208**. The one or more computer-readable storage media can include volatile and/or non-volatile storage components, such as optical, magnetic, organic or other memory or disc storage, which can be integrated in whole or in part with at least one of the one or more processors **208**. In some embodiments, the data storage **210** can be implemented using a single physical device (e.g., one optical, magnetic, organic or other memory or disc storage unit), while in other embodiments, the data storage **210** can be implemented using two or more physical devices.

(136) As noted, the data storage **210** can include computer-readable program instructions **212** and perhaps additional data, such as diagnostic data of the UAV **200**. As such, the data storage **210** may include program instructions **212** to perform or facilitate some or all of the UAV functionality described herein. For instance, in the illustrated embodiment, program instructions **212** include a navigation module **214** and a tether control module **216**.

(137) In an illustrative embodiment, IMU **202** may include both an accelerometer and a gyroscope, which may be used together to determine an orientation of the UAV **200**. In particular, the accelerometer can measure the orientation of the vehicle with respect to earth, while the gyroscope measures the rate of rotation around an axis. IMUs are commercially available in low-cost, low-power packages. For instance, an IMU **202** may take the form of or include a miniaturized MicroElectroMechanical System (MEMS) or a NanoElectroMechanical System (NEMS). Other types of IMUs may also be utilized.

(138) An IMU **202** may include other sensors, in addition to accelerometers and gyroscopes, which may help to better determine position and/or help to increase autonomy of the UAV **200**. Two examples of such sensors are magnetometers and pressure sensors. In some embodiments, a UAV may include a low-power, digital 3-axis magnetometer, which can be used to realize an orientation independent electronic compass for accurate heading information. However, other types of

magnetometers may be utilized as well. Other examples are also possible. Further, note that a UAV could include some or all of the above-described inertia sensors as separate components from an IMU.

(139) UAV **200** may also include a pressure sensor or barometer, which can be used to determine the altitude of the UAV **200**. Alternatively, other sensors, such as sonic altimeters or radar altimeters, can be used to provide an indication of altitude, which may help to improve the accuracy of and/or prevent drift of an IMU.

(140) In a further aspect, UAV **200** may include one or more sensors that allow the UAV to sense objects in the environment. For instance, in the illustrated embodiment, UAV **200** includes ultrasonic sensor(s) **204**. Ultrasonic sensor(s) **204** can determine the distance to an object by generating sound waves and determining the time interval between transmission of the wave and receiving the corresponding echo off an object. A typical application of an ultrasonic sensor for unmanned vehicles or IMUs is low-level altitude control and obstacle avoidance. An ultrasonic sensor can also be used for vehicles that need to hover at a certain height or need to be capable of detecting obstacles. Other systems can be used to determine, sense the presence of, and/or determine the distance to nearby objects, such as a light detection and ranging (LIDAR) system, laser detection and ranging (LADAR) system, and/or an infrared or forward-looking infrared (FLIR) system, among other possibilities.

(141) In some embodiments, UAV **200** may also include one or more imaging system(s). For example, one or more still and/or video cameras may be utilized by UAV **200** to capture image data from the UAV's environment. As a specific example, charge-coupled device (CCD) cameras or complementary metal-oxide-semiconductor (CMOS) cameras can be used with unmanned vehicles. Such imaging sensor(s) have numerous possible applications, such as obstacle avoidance, localization techniques, ground tracking for more accurate navigation (e.g., by applying optical flow techniques to images), video feedback, and/or image recognition and processing, among other possibilities.

(142) UAV **200** may also include a GPS receiver **206**. The GPS receiver **206** may be configured to provide data that is typical of well-known GPS systems, such as the GPS coordinates of the UAV **200**. Such GPS data may be utilized by the UAV **200** for various functions. As such, the UAV may use its GPS receiver **206** to help navigate to the caller's location, as indicated, at least in part, by the GPS coordinates provided by their mobile device. Other examples are also possible.

(143) The navigation module **214** may provide functionality that allows the UAV **200** to, e.g., move about its environment and reach a desired location. To do so, the navigation module **214** may control the altitude and/or direction of flight by controlling the mechanical features of the UAV that affect flight (e.g., its rudder(s), elevator(s), aileron(s), and/or the speed of its propeller(s)).

(144) In order to navigate the UAV **200** to a target location (e.g., a delivery location), the navigation module **214** may implement various navigation techniques, such as map-based navigation and localization-based navigation, for instance. With map-based navigation, the UAV **200** may be provided with a map of its environment, which may then be used to navigate to a particular location on the map. With localization-based navigation, the UAV **200** may be capable of navigating in an unknown environment using localization. Localization-based navigation may involve the UAV **200** building its own map of its environment and calculating its position within the map and/or the position of objects in the environment. For example, as a UAV **200** moves throughout its environment, the UAV **200** may continuously use localization to update its map of the environment. This continuous mapping process may be referred to as simultaneous localization and mapping (SLAM). Other navigation techniques may also be utilized.

(145) In some embodiments, the navigation module **214** may navigate using a technique that relies on waypoints. In particular, waypoints are sets of coordinates that identify points in physical space. For instance, an air-navigation waypoint may be defined by a certain latitude, longitude, and altitude. Accordingly, navigation module **214** may cause UAV **200** to move from waypoint to

waypoint, in order to ultimately travel to a final destination (e.g., a final waypoint in a sequence of waypoints).

(146) In a further aspect, the navigation module **214** and/or other components and systems of the UAV **200** may be configured for “localization” to more precisely navigate to the scene of a target location. More specifically, it may be desirable in certain situations for a UAV to be within a threshold distance of the target location where a payload **228** is being delivered by a UAV (e.g., within a few feet of the target destination). To this end, a UAV may use a two-tiered approach in which it uses a more-general location-determination technique to navigate to a general area that is associated with the target location, and then use a more-refined location-determination technique to identify and/or navigate to the target location within the general area.

(147) For example, the UAV **200** may navigate to the general area of a target destination where a payload **228** is being delivered using waypoints and/or map-based navigation. The UAV may then switch to a mode in which it utilizes a localization process to locate and travel to a more specific location. For instance, if the UAV **200** is to deliver a payload to a user's home, the UAV **200** may need to be substantially close to the target location in order to avoid delivery of the payload to undesired areas (e.g., onto a roof, into a pool, onto a neighbor's property, etc.). However, a GPS signal may only get the UAV **200** so far (e.g., within a block of the user's home). A more precise location-determination technique may then be used to find the specific target location.

(148) Various types of location-determination techniques may be used to accomplish localization of the target delivery location once the UAV **200** has navigated to the general area of the target delivery location. For instance, the UAV **200** may be equipped with one or more sensory systems, such as, for example, ultrasonic sensors **204**, infrared sensors (not shown), and/or other sensors, which may provide input that the navigation module **214** utilizes to navigate autonomously or semi-autonomously to the specific target location.

(149) As another example, once the UAV **200** reaches the general area of the target delivery location (or of a moving subject such as a person or their mobile device), the UAV **200** may switch to a “fly-by-wire” mode where it is controlled, at least in part, by a remote operator, who can navigate the UAV **200** to the specific target location. To this end, sensory data from the UAV **200** may be sent to the remote operator to assist them in navigating the UAV **200** to the specific location.

(150) As yet another example, the UAV **200** may include a module that is able to signal to a passer-by for assistance in either reaching the specific target delivery location; for example, the UAV **200** may display a visual message requesting such assistance in a graphic display, play an audio message or tone through speakers to indicate the need for such assistance, among other possibilities. Such a visual or audio message might indicate that assistance is needed in delivering the UAV **200** to a particular person or a particular location, and might provide information to assist the passer-by in delivering the UAV **200** to the person or location (e.g., a description or picture of the person or location, and/or the person or location's name), among other possibilities. Such a feature can be useful in a scenario in which the UAV is unable to use sensory functions or another location-determination technique to reach the specific target location. However, this feature is not limited to such scenarios.

(151) In some embodiments, once the UAV **200** arrives at the general area of a target delivery location, the UAV **200** may utilize a beacon from a user's remote device (e.g., the user's mobile phone) to locate the person. Such a beacon may take various forms. As an example, consider the scenario where a remote device, such as the mobile phone of a person who requested a UAV delivery, is able to send out directional signals (e.g., via an RF signal, a light signal and/or an audio signal). In this scenario, the UAV **200** may be configured to navigate by “sourcing” such directional signals—in other words, by determining where the signal is strongest and navigating accordingly. As another example, a mobile device can emit a frequency, either in the human range or outside the human range, and the UAV **200** can listen for that frequency and navigate accordingly. As a related

example, if the UAV **200** is listening for spoken commands, then the UAV **200** could utilize spoken statements, such as “I'm over here!” to source the specific location of the person requesting delivery of a payload.

(152) In an alternative arrangement, a navigation module may be implemented at a remote computing device, which communicates wirelessly with the UAV **200**. The remote computing device may receive data indicating the operational state of the UAV **200**, sensor data from the UAV **200** that allows it to assess the environmental conditions being experienced by the UAV **200**, and/or location information for the UAV **200**. Provided with such information, the remote computing device may determine altitudinal and/or directional adjustments that should be made by the UAV **200** and/or may determine how the UAV **200** should adjust its mechanical features (e.g., its rudder(s), elevator(s), aileron(s), and/or the speed of its propeller(s)) in order to effectuate such movements. The remote computing system may then communicate such adjustments to the UAV **200** so it can move in the determined manner.

(153) In a further aspect, the UAV **200** includes one or more communication systems **218**. The communications systems **218** may include one or more wireless interfaces and/or one or more wireline interfaces, which allow the UAV **200** to communicate via one or more networks. Such wireless interfaces may provide for communication under one or more wireless communication protocols, such as Bluetooth, WiFi (e.g., an IEEE 802.11 protocol), Long-Term Evolution (LTE), WiMAX (e.g., an IEEE 802.16 standard), a radio-frequency ID (RFID) protocol, near-field communication (NFC), and/or other wireless communication protocols. Such wireline interfaces may include an Ethernet interface, a Universal Serial Bus (USB) interface, or similar interface to communicate via a wire, a twisted pair of wires, a coaxial cable, an optical link, a fiber-optic link, or other physical connection to a wireline network.

(154) In some embodiments, a UAV **200** may include communication systems **218** that allow for both short-range communication and long-range communication. For example, the UAV **200** may be configured for short-range communications using Bluetooth and for long-range communications under a CDMA protocol. In such an embodiment, the UAV **200** may be configured to function as a “hot spot;” or in other words, as a gateway or proxy between a remote support device and one or more data networks, such as a cellular network and/or the Internet. Configured as such, the UAV **200** may facilitate data communications that the remote support device would otherwise be unable to perform by itself.

(155) For example, the UAV **200** may provide a WiFi connection to a remote device, and serve as a proxy or gateway to a cellular service provider's data network, which the UAV might connect to under an LTE or a 3G protocol, for instance. The UAV **200** could also serve as a proxy or gateway to a high-altitude balloon network, a satellite network, or a combination of these networks, among others, which a remote device might not be able to otherwise access.

(156) In a further aspect, the UAV **200** may include power system(s) **220**. The power system **220** may include one or more batteries for providing power to the UAV **200**. In one example, the one or more batteries may be rechargeable and each battery may be recharged via a wired connection between the battery and a power supply and/or via a wireless charging system, such as an inductive charging system that applies an external time-varying magnetic field to an internal battery.

(157) In a further aspect, the power systems **220** of UAV **200** a power interface for electronically coupling to an external AC power source, and an AC/DC converter coupled to the power interface and operable to convert alternating current to direct current that charges the UAV's battery or batteries. For instance, the power interface may include a power jack or other electric coupling for connecting to a 110V, 120V, 220V, or 240V AC power source. Such a power system may facilitate a recipient-assisted recharging process, where a recipient can connect the UAV to a standard power source at a delivery location, such as the recipient's home or office. Additionally or alternatively, power systems **220** could include an inductive charging interface, such that recipient-assisted recharging can be accomplished wirelessly via an inductive charging system installed or otherwise

available at the delivery location.

(158) The UAV **200** may employ various systems and configurations in order to transport and deliver a payload **228**. In some implementations, the payload **228** of a given UAV **200** may include or take the form of a "package" designed to transport various goods to a target delivery location. For example, the UAV **200** can include a compartment, in which an item or items may be transported. Such a package may contain one or more food items, purchased goods, medical items, or any other object(s) having a size and weight suitable to be transported between two locations by the UAV. In some embodiments, a payload **228** may simply be the one or more items that are being delivered (e.g., without any package housing the items). And, in some embodiments, the items being delivered, the container or package in which the items are transported, and/or other components may all be considered to be part of the payload.

(159) In some embodiments, the payload **228** may be attached to the UAV and located substantially outside of the UAV during some or all of a flight by the UAV. For example, the package may be tethered or otherwise releasably attached below the UAV during flight to a target location. In an embodiment where a package carries goods below the UAV, the package may include various features that protect its contents from the environment, reduce aerodynamic drag on the system, and prevent the contents of the package from shifting during UAV flight.

(160) For instance, when the payload **228** takes the form of a package for transporting items, the package may include an outer shell constructed of water-resistant cardboard, plastic, or any other lightweight and water-resistant material. Further, in order to reduce drag, the package may feature smooth surfaces with a pointed front that reduces the frontal cross-sectional area. Further, the sides of the package may taper from a wide bottom to a narrow top, which allows the package to serve as a narrow pylon that reduces interference effects on the wing(s) of the UAV. This may move some of the frontal area and volume of the package away from the wing(s) of the UAV, thereby preventing the reduction of lift on the wing(s) caused by the package. Yet further, in some embodiments, the outer shell of the package may be constructed from a single sheet of material in order to reduce air gaps or extra material, both of which may increase drag on the system. Additionally or alternatively, the package may include a stabilizer to dampen package flutter. This reduction in flutter may allow the package to have a less rigid connection to the UAV and may cause the contents of the package to shift less during flight.

(161) In order to deliver the payload, the UAV may include a tether system **221**, which may be controlled by the tether control module **216** in order to lower the payload **228** to the ground while the UAV hovers above. The tether system **221** may include a tether, which is couplable to a payload **228** (e.g., a package). The tether may be wound on a spool that is coupled to a motor of the UAV (although passive implementations, without a motor, are also possible). The motor may be a DC motor (e.g., a servo motor) that can be actively controlled by a speed controller, although other motor configurations are possible. In some embodiments, the tether control module **216** can control the speed controller to cause the motor to rotate the spool, thereby unwinding or retracting the tether and lowering or raising the payload coupling apparatus. In practice, a speed controller may output a desired operating rate (e.g., a desired RPM) for the spool, which may correspond to the speed at which the tether system should lower the payload towards the ground. The motor may then rotate the spool so that it maintains the desired operating rate (or within some allowable range of operating rates).

(162) In order to control the motor via a speed controller, the tether control module **216** may receive data from a speed sensor (e.g., an encoder) configured to convert a mechanical position to a representative analog or digital signal. In particular, the speed sensor may include a rotary encoder that may provide information related to rotary position (and/or rotary movement) of a shaft of the motor or the spool coupled to the motor, among other possibilities. Moreover, the speed sensor may take the form of an absolute encoder and/or an incremental encoder, among others. So in an example implementation, as the motor causes rotation of the spool, a rotary encoder may be used to

measure this rotation. In doing so, the rotary encoder may be used to convert a rotary position to an analog or digital electronic signal used by the tether control module **216** to determine the amount of rotation of the spool from a fixed reference angle and/or to an analog or digital electronic signal that is representative of a new rotary position, among other options. Other examples are also possible.

(163) In some embodiments, a payload coupling component (e.g., a hook or another type of coupling component) can be configured to secure the payload **228** while being lowered from the UAV by the tether. The coupling apparatus or component can be further configured to release the payload **228** upon reaching ground level via electrical or electro-mechanical features of the coupling component. The payload coupling component can then be retracted to the UAV by reeling in the tether using the motor.

(164) In some implementations, the payload **228** may be passively released once it is lowered to the ground. For example, a payload coupling component may provide a passive release mechanism, such as one or more swing arms adapted to retract into and extend from a housing. An extended swing arm may form a hook on which the payload **228** may be attached. Upon lowering the release mechanism and the payload **228** to the ground via a tether, a gravitational force as well as a downward inertial force on the release mechanism may cause the payload **228** to detach from the hook allowing the release mechanism to be raised upwards toward the UAV. The release mechanism may further include a spring mechanism that biases the swing arm to retract into the housing when there are no other external forces on the swing arm. For instance, a spring may exert a force on the swing arm that pushes or pulls the swing arm toward the housing such that the swing arm retracts into the housing once the weight of the payload **228** no longer forces the swing arm to extend from the housing. Retracting the swing arm into the housing may reduce the likelihood of the release mechanism snagging the payload **228** or other nearby objects when raising the release mechanism toward the UAV upon delivery of the payload **228**.

(165) In another implementation, a payload coupling component may include a hook feature that passively releases the payload when the payload contacts the ground. For example, the payload coupling component may take the form of or include a hook feature that is sized and shaped to interact with a corresponding attachment feature (e.g., a handle or hole) on a payload taking the form of a container or tote. The hook may be inserted into the handle or hole of the payload container, such that the weight of the payload keeps the payload container secured to the hook feature during flight. However, the hook feature and payload container may be designed such that when the container contacts the ground and is supported from below, the hook feature slides out of the container's attachment feature, thereby passively releasing the payload container. Other passive release configurations are also possible.

(166) Active payload release mechanisms are also possible. For example, sensors such as a barometric pressure based altimeter and/or accelerometers may help to detect the position of the release mechanism (and the payload) relative to the ground. Data from the sensors can be communicated back to the UAV and/or a control system over a wireless link and used to help in determining when the release mechanism has reached ground level (e.g., by detecting a measurement with the accelerometer that is characteristic of ground impact). In other examples, the UAV may determine that the payload has reached the ground based on a weight sensor detecting a threshold low downward force on the tether and/or based on a threshold low measurement of power drawn by the winch when lowering the payload.

(167) Other systems and techniques for delivering a payload, in addition or in the alternative to a tethered delivery system are also possible. For example, a UAV **200** could include an air-bag drop system or a parachute drop system. Alternatively, a UAV **200** carrying a payload could simply land on the ground at a delivery location. Other examples are also possible.

(168) In some arrangements, a UAV may not include a tether system **221**. For example, a UAV could include an internal compartment or bay in which the UAV could hold items during transport.

Such a compartment could be configured as a top-loading, side-loading, and/or bottom-loading chamber. The UAV may include electrical and/or mechanical means (e.g., doors) that allow the interior compartment in the UAV to be opened and closed. Accordingly, the UAV may open the compartment in various circumstances, such as: (a) when picking up an item for delivery at an item source location, such that the item can be placed in the UAV for delivery, (b) upon arriving at a delivery location, such that the recipient can place an item for return into the UAV, and/or (c) in other circumstances. Further, it is also contemplated that other non-tethered mechanisms for securing payload items to a UAV are also possible, such as various fasteners for securing items to the UAV housing, among other possibilities. Yet further, a UAV may include an internal compartment for transporting items and/or other non-tethered mechanisms for securing payload items, in addition or in the alternative to a tether system **221**.
(169) The UAV **200** can include a package identification device **230** that can be used to identify payload **228**. Within examples, the package identification device **230** can be arranged on a surface of the UAV **200** that has a direct view of the payload **228**. For instance, the package identification device **230** can be arranged on a surface of a payload compartment (see, e.g., compartments **506** and **604** in FIGS. **5** and **6**, respectively). Other arrangements are possible.
(170) Within examples, the package identification device **230** can be a sensor or a scanner that employs various technologies to interact with the payload **228** in order to identify the payload **228**. For instance, the package identification device **230** can employ one or more wireless communication protocols, such as Bluetooth, WiFi (e.g., an IEEE 802.11 protocol), Long-Term Evolution (LTE), WiMAX (e.g., an IEEE 802.16 standard), a radio-frequency ID (RFID) protocol, near-field communication (NFC), and/or other wireless communication protocols. Additionally and/or alternatively, the package identification device **230** can employ various scanning technologies such as a 1-D or 2-D barcode scanner. Additionally and/or alternatively, the package identification device **230** can employ various image-capturing technologies such as cameras. Additionally and/or alternatively, the package identification device **230** can employ various text recognition software that can read identifying text (e.g., printed or handwritten) on the package.
(171) UAV systems may be implemented in order to provide various UAV-related services. In particular, UAVs may be provided at a number of different launch sites that may be in communication with regional and/or central control systems. Such a distributed UAV system may allow UAVs to be quickly deployed to provide services across a large geographic area (e.g., that is much larger than the flight range of any single UAV). For example, UAVs capable of carrying payloads may be distributed at a number of launch sites across a large geographic area (possibly even throughout an entire country, or even worldwide), in order to provide on-demand transport of various items to locations throughout the geographic area. FIG. **3**A is a simplified block diagram illustrating a distributed UAV system **300**, according to an example embodiment.
(172) In the illustrative UAV system **300** shown in FIG. **3**, an access system **302** may allow for interaction with, control of, and/or utilization of a network of UAVs **304**. In some embodiments, an access system **302** may be a computing system that allows for human-controlled dispatch of UAVs **304**. As such, the control system may include or otherwise provide a user interface through which a user can access and/or control the UAVs **304**.
(173) In some embodiments, dispatch of the UAVs **304** may additionally or alternatively be accomplished via one or more automated processes. For instance, the access system **302** may dispatch one of the UAVs **304** to transport a payload to a target location, and the UAV may autonomously navigate to the target location by utilizing various on-board sensors, such as a GPS receiver and/or other various navigational sensors.
(174) Further, the access system **302** may provide for remote operation of a UAV. For instance, the access system **302** may allow an operator to control the flight of a UAV via its user interface. As a specific example, an operator may use the access system **302** to dispatch a UAV **304** to a target location. The UAV **304** may then autonomously navigate to the general area of the target location.

At this point, the operator may use the access system **302** to take control of the UAV **304** and navigate the UAV to the target location (e.g., to a particular person to whom a payload is being transported). Other examples of remote operation of a UAV are also possible.

(175) In an illustrative embodiment, the UAVs **304** may take various forms. For example, each of the UAVs **304** may be a UAV such as those illustrated in FIGS. **1**A-**1**E. However, UAV system **300** may also utilize other types of UAVs without departing from the scope of the invention. In some implementations, all of the UAVs **304** may be of the same or a similar configuration. However, in other implementations, the UAVs **304** may include a number of different types of UAVs. For instance, the UAVs **304** may include a number of types of UAVs, with each type of UAV being configured for a different type or types of payload delivery capabilities.

(176) The UAV system **300** may further include a remote device **306**, which may take various forms. Generally, the remote device **306** may be any device through which a direct or indirect request to dispatch a UAV can be made. (Note that an indirect request may involve any communication that may be responded to by dispatching a UAV, such as requesting a package delivery). In an example embodiment, the remote device **306** may be a mobile phone, tablet computer, laptop computer, personal computer, or any network-connected computing device. Further, in some instances, the remote device **306** may not be a computing device. As an example, a standard telephone, which allows for communication via plain old telephone service (POTS), may serve as the remote device **306**. Other types of remote devices are also possible.

(177) Further, the remote device **306** may be configured to communicate with access system **302** via one or more types of communication network(s) **308**. For example, the remote device **306** may communicate with the access system **302** (or a human operator of the access system **302**) by communicating over a POTS network, a cellular network, and/or a data network such as the Internet. Other types of networks may also be utilized.

(178) In some embodiments, the remote device **306** may be configured to allow a user to request delivery of one or more items to a desired location. For example, a user could request UAV delivery of a package to their home via their mobile phone, tablet, or laptop. As another example, a user could request dynamic delivery to wherever they are located at the time of delivery. To provide such dynamic delivery, the UAV system **300** may receive location information (e.g., GPS coordinates, etc.) from the user's mobile phone, or any other device on the user's person, such that a UAV can navigate to the user's location (as indicated by their mobile phone).

(179) In an illustrative arrangement, the central dispatch system **310** may be a server or group of servers, which is configured to receive dispatch messages requests and/or dispatch instructions from the access system **302**. Such dispatch messages may request or instruct the central dispatch system **310** to coordinate the deployment of UAVs to various target locations. The central dispatch system **310** may be further configured to route such requests or instructions to one or more local dispatch systems **312**. To provide such functionality, the central dispatch system **310** may communicate with the access system **302** via a data network, such as the Internet or a private network that is established for communications between access systems and automated dispatch systems.

(180) In the illustrated configuration, the central dispatch system **310** may be configured to coordinate the dispatch of UAVs **304** from a number of different local dispatch systems **312**. As such, the central dispatch system **310** may keep track of which UAVs **304** are located at which local dispatch systems **312**, which UAVs **304** are currently available for deployment, and/or which services or operations each of the UAVs **304** is configured for (in the event that a UAV fleet includes multiple types of UAVs configured for different services and/or operations). Additionally or alternatively, each local dispatch system **312** may be configured to track which of its associated UAVs **304** are currently available for deployment and/or are currently in the midst of item transport.

(181) In some cases, when the central dispatch system **310** receives a request for UAV-related

service (e.g., transport of an item) from the access system **302**, the central dispatch system **310** may select a specific UAV **304** to dispatch. The central dispatch system **310** may accordingly instruct the local dispatch system **312** that is associated with the selected UAV to dispatch the selected UAV. The local dispatch system **312** may then operate its associated deployment system **314** to launch the selected UAV. In other cases, the central dispatch system **310** may forward a request for a UAV-related service to a local dispatch system **312** that is near the location where the support is requested and leave the selection of a particular UAV **304** to the local dispatch system **312**.
(182) In an example configuration, the local dispatch system **312** may be implemented as a computing system at the same location as the deployment system(s) **314** that it controls. For example, the local dispatch system **312** may be implemented by a computing system installed at a building, such as a warehouse, where the deployment system(s) **314** and UAV(s) **304** that are associated with the particular local dispatch system **312** are also located. In other embodiments, the local dispatch system **312** may be implemented at a location that is remote to its associated deployment system(s) **314** and UAV(s) **304**.
(183) Numerous variations on and alternatives to the illustrated configuration of the UAV system **300** are possible. For example, in some embodiments, a user of the remote device **306** could request delivery of a package directly from the central dispatch system **310**. To do so, an application may be implemented on the remote device **306** that allows the user to provide information regarding a requested delivery, and generate and send a data message to request that the UAV system **300** provide the delivery. In such an embodiment, the central dispatch system **310** may include automated functionality to handle requests that are generated by such an application, evaluate such requests, and, if appropriate, coordinate with an appropriate local dispatch system **312** to deploy a UAV.
(184) Further, some or all of the functionality that is attributed herein to the central dispatch system **310**, the local dispatch system(s) **312**, the access system **302**, and/or the deployment system(s) **314** may be combined in a single system, implemented in a more complex system, and/or redistributed among the central dispatch system **310**, the local dispatch system(s) **312**, the access system **302**, and/or the deployment system(s) **314** in various ways.
(185) Yet further, while each local dispatch system **312** is shown as having two associated deployment systems **314**, a given local dispatch system **312** may alternatively have more or fewer associated deployment systems **314**. Similarly, while the central dispatch system **310** is shown as being in communication with two local dispatch systems **312**, the central dispatch system **310** may alternatively be in communication with more or fewer local dispatch systems **312**.
(186) In a further aspect, the deployment systems **314** may take various forms. In general, the deployment systems **314** may take the form of or include systems for physically launching one or more of the UAVs **304**. Such launch systems may include features that provide for an automated UAV launch and/or features that allow for a human-assisted UAV launch. Further, the deployment systems **314** may each be configured to launch one particular UAV **304**, or to launch multiple UAVs **304**.
(187) The deployment systems **314** may further be configured to provide additional functions, including for example, diagnostic-related functions such as verifying system functionality of the UAV, verifying functionality of devices that are housed within a UAV (e.g., a payload delivery apparatus), and/or maintaining devices or other items that are housed in the UAV (e.g., by monitoring a status of a payload such as its temperature, weight, etc.).
(188) In some embodiments, the deployment systems **314** and their corresponding UAVs **304** (and possibly associated local dispatch systems **312**) may be strategically distributed throughout an area such as a city. For example, the deployment systems **314** may be strategically distributed such that each deployment system **314** is proximate to one or more payload pickup locations (e.g., near a restaurant, store, or warehouse). However, the deployment systems **314** (and possibly the local dispatch systems **312**) may be distributed in other ways, depending upon the particular

implementation. As an additional example, kiosks that allow users to transport packages via UAVs may be installed in various locations. Such kiosks may include UAV launch systems, and may allow a user to provide their package for loading onto a UAV and pay for UAV shipping services, among other possibilities. Other examples are also possible.
(189) In a further aspect, the UAV system **300** may include or have access to a user-account database **316**. The user-account database **316** may include data for a number of user accounts, and which are each associated with one or more persons. For a given user account, the user-account database **316** may include data related to or useful in providing UAV-related services. Typically, the user data associated with each user account is optionally provided by an associated user and/or is collected with the associated user's permission.
(190) Further, in some embodiments, a person may be required to register for a user account with the UAV system **300**, if they wish to be provided with UAV-related services by the UAVs **304** from UAV system **300**. As such, the user-account database **316** may include authorization information for a given user account (e.g., a username and password), and/or other information that may be used to authorize access to a user account.
(191) In some embodiments, a person may associate one or more of their devices with their user account, such that they can access the services of UAV system **300**. For example, when a person uses an associated mobile phone, e.g., to place a call to an operator of the access system **302** or send a message requesting a UAV-related service to a dispatch system, the phone may be identified via a unique device identification number, and the call or message may then be attributed to the associated user account. Other examples are also possible.
(192) FIGS. **4**A, **4**B, and **4**C show a UAV **400** that includes a payload delivery system **410** (could also be referred to as a payload delivery apparatus), according to an example embodiment. As shown, payload delivery system **410** for UAV **400** includes a tether **402** coupled to a spool **404**, a payload latch **406**, and a payload **408** coupled to the tether **402** via a payload coupling apparatus **412**. The payload latch **406** can function to alternately secure payload **408** and release the payload **408** upon delivery. For instance, as shown, the payload latch **406** may take the form of one or more pins that can engage the payload coupling apparatus **412** (e.g., by sliding into one or more receiving slots in the payload coupling apparatus **412**). Inserting the pins of the payload latch **406** into the payload coupling apparatus **412** may secure the payload coupling apparatus **412** within a receptacle **414** on the underside of the UAV **400**, thereby preventing the payload **408** from being lowered from the UAV **400**. In some embodiments, the payload latch **406** may be arranged to engage the spool **404** or the payload **408** rather than the payload coupling apparatus **412** in order to prevent the payload **408** from lowering. In other embodiments, the UAV **400** may not include the payload latch **406**, and the payload delivery apparatus may be coupled directly to the UAV **400**.
(193) In some embodiments, the spool **404** can function to unwind the tether **402** such that the payload **408** can be lowered to the ground with the tether **402** and the payload coupling apparatus **412** from UAV **400**. The payload **408** may itself be an item for delivery, and may be housed within (or otherwise incorporate) a parcel, container, or other structure that is configured to interface with the payload latch **406**. In practice, the payload delivery system **410** of UAV **400** may function to autonomously lower payload **408** to the ground in a controlled manner to facilitate delivery of the payload **408** on the ground while the UAV **400** hovers above.
(194) As shown in FIG. **4**A, the payload latch **406** may be in a closed position (e.g., pins engaging the payload coupling apparatus **412**) to hold the payload **408** against or close to the bottom of the UAV **400**, or even partially or completely inside the UAV **400**, during flight from a launch site to a target location **420**. The target location **420** may be a point in space directly above a desired delivery location. Then, when the UAV **400** reaches the target location **420**, the UAV's control system (e.g., the tether control module **216** of FIG. **2**) may toggle the payload latch **406** to an open position (e.g., disengaging the pins from the payload coupling apparatus **412**), thereby allowing the payload **408** to be lowered from the UAV **400**. The control system may further operate the spool

404 (e.g., by controlling the motor **222** of FIG. **2**) such that the payload **408**, secured to the tether **402** by a payload coupling apparatus **412**, is lowered to the ground, as shown in FIG. **4**B.
(195) Once the payload **408** reaches the ground, the control system may continue operating the spool **404** to lower the tether **402**, causing over-run of the tether **402**. During over-run of the tether **402**, the payload coupling apparatus **412** may continue to lower as the payload **408** remains stationary on the ground. The downward momentum and/or gravitational forces on the payload coupling apparatus **412** may cause the payload **408** to detach from the payload coupling apparatus **412** (e.g., by sliding off a hook of the payload coupling apparatus **412**). After releasing payload **408**, the control system may operate the spool **404** to retract the tether **402** and the payload coupling apparatus **412** toward the UAV **400**. Once the payload coupling apparatus reaches or nears the UAV **400**, the control system may operate the spool **404** to pull the payload coupling apparatus **412** into the receptacle **414**, and the control system may toggle the payload latch **406** to the closed position, as shown in FIG. **4**C.
(196) In some embodiments, when lowering the payload **408** from the UAV **400**, the control system may detect when the payload **408** and/or the payload coupling apparatus **412** has been lowered to be at or near the ground based on an unwound length of the tether **402** from the spool **404**. Similar techniques may be used to determine when the payload coupling apparatus **412** is at or near the UAV **400** when retracting the tether **402**. As noted above, the UAV **400** may include an encoder for providing data indicative of the rotation of the spool **404**. Based on data from the encoder, the control system may determine how many rotations the spool **404** has undergone and, based on the number of rotations, determine a length of the tether **402** that is unwound from the spool **404**. For instance, the control system may determine an unwound length of the tether **402** by multiplying the number of rotations of the spool **404** by the circumference of the tether **402** wrapped around the spool **404**. In some embodiments, such as when the spool **404** is narrow or when the tether **402** has a large diameter, the circumference of the tether **402** on the spool **404** may vary as the tether **402** winds or unwinds from the tether, and so the control system may be configured to account for these variations when determining the unwound tether length.
(197) In other embodiments, the control system may use various types of data, and various techniques, to determine when the payload **408** and/or payload coupling apparatus **412** have lowered to be at or near the ground. Further, the data that is used to determine when the payload **408** is at or near the ground may be provided by sensors on UAV **400**, sensors on the payload coupling apparatus **412**, and/or other data sources that provide data to the control system.
(198) In some embodiments, the control system itself may be situated on the payload coupling apparatus **412** and/or on the UAV **400**. For example, the payload coupling apparatus **412** may include logic module(s) implemented via hardware, software, and/or firmware that cause the UAV **400** to function as described herein, and the UAV **400** may include logic module(s) that communicate with the payload coupling apparatus **412** to cause the UAV **400** to perform functions described herein.
(199) FIG. **5**A shows a perspective view of a payload delivery apparatus **500** including payload **510**, according to an example embodiment. The payload delivery apparatus **500** is positioned within a fuselage of a UAV (not shown) and includes a winch **514** powered by motor **512**, and a tether **502** spooled onto winch **514**. The tether **502** is attached to a payload coupling apparatus or payload retriever **800** positioned within a payload coupling apparatus receptacle **516** positioned within the fuselage of the UAV (not shown). A payload **510** is secured to the payload coupling apparatus **800**. In this embodiment a top portion **517** of payload **510** is secured within the fuselage of the UAV. A locking pin **570** is shown extending through handle **511** attached to payload **510** to positively secure the payload beneath the UAV during high speed flight.
(200) FIG. **5**B is a cross-sectional side view of payload delivery apparatus **500** and payload **510** shown in FIG. **5**A. In this view, the payload coupling apparatus is shown tightly positioned with the payload coupling apparatus receptacle **516**. Tether **502** extends from winch **514** and is attached to

the top of payload coupling apparatus **800**. Top portion **517** of payload **510** is shown positioned within the fuselage of the UAV (not shown) along with handle **511**.

(201) FIG. **5**C is a side view of payload delivery apparatus **500** and payload **510** shown in FIGS. **5**A and **5**B. The top portion **517** of payload **510** is shown positioned within the fuselage of the UAV. Winch **514** has been used to wind in tether **502** to position the payload coupling apparatus within payload coupling apparatus receptacle **516**. FIGS. **5**A-C disclose payload **510** taking the shape of an aerodynamic hexagonally-shaped tote, where the base and side walls are six-sided hexagons and the tote includes generally pointed front and rear surfaces formed at the intersections of the side walls and base of the tote providing an aerodynamic shape.

(202) FIG. **6**A is a perspective view of payload coupling apparatus **800**, which may also be referred to as a payload retriever, according to an example embodiment. Payload coupling apparatus **800** includes tether mounting point **802**, and a slot **808** to position a handle of a payload handle in. Lower lip, or hook, **806** is positioned beneath slot **808**. Also included is an outer protrusion **804** having helical cam surfaces **804***a* and **804***b* that are adapted to mate with corresponding cam mating surfaces within a payload coupling apparatus receptacle positioned with a fuselage of a UAV.

(203) FIG. **6**B is a side view of payload coupling apparatus **800** shown in FIG. **6**A. Slot **808** is shown positioned above lower lip, or hook, **806**. As shown lower lip or hook **806** has an outer surface **806***a* that is undercut such that it does not extend as far outwardly as an outer surface above slot **805** so that the lower lip or hook **806** will not reengage with the handle of the payload after it has been decoupled, or will not get engaged with power lines or tree branches during retrieval to the UAV.

(204) FIG. **6**C is a front view of payload coupling apparatus **800** shown in FIGS. **6**A and **6**B. Lower lip or hook **806** is shown positioned beneath slot **808** that is adapted for securing a handle of a payload.

(205) FIG. **7** is a perspective view of payload coupling apparatus **800** shown in FIGS. **6**A-**6**C, prior to insertion into a payload coupling apparatus receptacle **516** positioned in the fuselage **550** of a UAV. As noted previously, payload coupling apparatus **800** includes a slot **808** positioned above lower lip or hook **806**, adapted to receive a handle of a payload. The fuselage **550** of the payload delivery system **500** includes a payload coupling apparatus receptacle **516** positioned within the fuselage **550** of the UAV. The payload coupling apparatus **800** includes an outer protrusion **810** have helical cammed surfaces **810***a* and **810***b* that meet in a rounded apex. The helical cammed surfaces **810***a* and **810***b* are adapted to mate with surfaces **530***a* and **530***b* of an inward protrusion **530** positioned within the payload coupling apparatus receptacle **516** positioned within fuselage **550** of the UAV. Also included is a longitudinal recessed restraint slot **540** positioned within the fuselage **550** of the UAV that is adapted to receive and restrain a top portion of a payload (not shown). As the payload coupling apparatus **800** is pulled into to the payload coupling apparatus receptacle **516**, the cammed surfaces **810***a* and **810***b* of outer protrusion **810** engage with the cammed surfaces **530***a* and **530***b* within the payload coupling apparatus receptacle **516** and the payload coupling apparatus **800** is rotated into a desired alignment within the fuselage **550** of the UAV.

(206) FIG. **8** is another perspective view of an opposite side of payload coupling apparatus **800** shown in FIGS. **6**A-**6**C, prior to insertion into a payload coupling apparatus receptacle **516** positioned in the fuselage **550** of a UAV. As shown, payload coupling apparatus **800** include a lower lip or hook **806**. An outer protrusion **804** is shown extending outwardly from the payload coupling apparatus having helical cammed surfaces **804***a* and **804***b* adapted to engage and mate with cammed surfaces **530***a* and **530***b* of inner protrusion **530** positioned within payload coupling apparatus receptacle **516** positioned within fuselage **550** of payload delivery system **500**. It should be noted that the cammed surfaces **804***a* and **804***b* meet at a sharp apex, which is asymmetrical with the rounded or blunt apex of cammed surfaces **810***a* and **810***b* shown in FIG. **7**. In this manner, the

rounded or blunt apex of cammed surfaces **810***a* and **810***b* prevent possible jamming of the payload coupling apparatus **800** as the cammed surfaces engage the cammed surfaces **530***a* and **530***b* positioned within the payload coupling apparatus receptacle **516** positioned within fuselage **550** of the UAV. In particular, cammed surfaces **804***a* and **804***b* are positioned slightly higher than the rounded or blunt apex of cammed surfaces **810***a* and **810***b*. As a result, the sharper tip of cammed surfaces **804***a* and **804***b* engages the cammed surfaces **530***a* and **530***b* within the payload coupling apparatus receptacle **516** positioned within the fuselage **550** of payload delivery system **500**, thereby initiating rotation of the payload coupling apparatus **800** slightly before the rounded or blunt apex of cammed surfaces **810***a* and **810***b* engage the corresponding cammed surfaces within the payload coupling apparatus receptacle **516**. In this manner, the case where both apexes (or tips) of the cammed surfaces on the payload coupling apparatus end up on the same side of the receiving cams within the payload coupling apparatus receptacle is prevented. This scenario results in a prevention of the jamming of the payload coupling apparatus within the receptacle.

(207) FIG. **9** shows a perspective view of a recessed restraint slot and payload coupling apparatus receptacle positioned in a fuselage of a UAV. In particular, payload delivery system **500** includes a fuselage **550** having a payload coupling apparatus receptacle **516** therein that includes inward protrusion **530** having cammed surfaces **530***a* and **530***b* that are adapted to mate with corresponding cammed surfaces on a payload coupling apparatus (not shown). Also included is a longitudinally extending recessed restrained slot **540** into which a top portion of a payload is adapted to be positioned and secured within the fuselage **550**.

(208) FIG. **10**A shows a side view of a payload delivery apparatus **500** with a handle **511** of payload **510** secured within a payload coupling apparatus **800** as the payload **510** moves downwardly prior to touching down for delivery. Prior to payload touchdown, the handle **511** of payload **510** includes a hole **513** through which a lower lip or hook of payload coupling apparatus **800** extends. The handle sits within a slot of the payload coupling apparatus **800** that is suspended from tether **502** of payload delivery system **500** during descent of the payload **510** to a landing site.

(209) FIG. **10**B shows a side view of payload delivery apparatus **500** after payload **510** has landed on the ground showing payload coupling apparatus **800** decoupled from handle **511** of payload **510**. Once the payload **510** touches the ground, the payload coupling apparatus **800** continues to move downwardly (as the winch further unwinds) through inertia or gravity and decouples the lower lip or hook **808** of the payload coupling apparatus **800** from handle **511** of payload **510**. The payload coupling apparatus **800** remains suspended from tether **502**, and can be winched back up to the payload coupling receptacle of the UAV.

(210) FIG. **10**C shows a side view of payload delivery apparatus **500** with payload coupling apparatus **800** moving away from handle **511** of payload **510**. Here the payload coupling apparatus **800** is completely separated from the hole **513** of handle **511** of payload **510**. Tether **502** may be used to winch the payload coupling apparatus back to the payload coupling apparatus receptacle positioned in the fuselage of the UAV.

(211) FIG. **11**A is a side view of handle **511** of payload **510**. The handle **511** includes an aperture **513** through which the lower lip or hook of a payload coupling apparatus extends through to suspend the payload during delivery, or for retrieval. The handle **511** includes a lower portion **515** that is secured to the top portion of a payload. Also included are holes **524** and **526** through which locking pins positioned within the fuselage of a UAV, may extend to secure the handle and payload in a secure position during high speed forward flight to a delivery location. In addition, holes **524** and **526** are also designed for pins of a payload holder to extend therethrough to hold the payload in position for retrieval on a payload retrieval apparatus. The handle may be comprised of a thin, flexible plastic material that is flexible and provides sufficient strength to suspend the payload beneath a UAV during forward flight to a delivery site, and during delivery and/or retrieval of a payload. In practice, the handle may be bent to position the handle within a slot of a payload coupling apparatus. The handle **511** also has sufficient strength to withstand the torque during

rotation of the payload coupling apparatus into the desired orientation within the payload coupling apparatus receptacle, and rotation of the top portion of the payload into position with the recessed restraint slot.

(212) FIG. **11**B is a side view of handle **511**′ of payload **510**. The handle **511**′ includes an aperture **513** through which the lower lip or hook of a payload coupling apparatus extends through to suspend the payload during delivery, or for retrieval. The handle **511**′ includes a lower portion **515** that is secured to the top portion of a payload. Also included are magnets **524**′ and **526**′ adapted for magnetic engagement with corresponding magnets (or a metal) of a payload holder to secure the payload to the payload holder in position for retrieval on a payload retrieval apparatus. In some examples, magnets **524**′ and **526**′ are provided on a handle (e.g., handle **511** or **511**′) in place of holes **524** and **526**. In other examples, magnets **524**′ and **526**′ are provided in addition to holes **524** and **526**.

(213) FIG. **12** shows a pair of pins **570**, **572** extending through holes **524** and **526** in handle **511** of payload **510** to secure the handle **511** and top portion of payload **510** within the fuselage of a UAV, or to secure payload **510** to a payload holder of a payload retrieval apparatus. In this manner, the handle **511** and payload **510** may be secured within the fuselage of a UAV, or to a payload holder of a payload retrieval apparatus. In this embodiment, the pins **570** and **572** have a conical shape so that they pull the package up slightly or at least remove any downward slack present. In some embodiments the pins **570** and **572** may completely plug the holes **524** and **526** of the handle **511** of payload **510**, to provide a secure attachment of the handle and top portion of the payload within the fuselage of the UAV, or to secure the payload to a payload retrieval apparatus. Although the pins are shown as conical, in other applications they may have other geometries, such as a cylindrical geometry.

(214) FIGS. **13**A and **13**B show various views of payload coupling apparatus or payload retriever **800**′ which is a variation of payload coupling apparatus **800** described above. Payload coupling apparatus **800**′ includes the same exterior features as payload coupling apparatus **800**. However, in payload coupling apparatus **800**′, lower lip or hook **806**′ is extendable and retractable. As shown in FIG. **13**A, payload coupling **800**′ is in a retracted state where end **806***a*′ of lip or hook **806**′ is positioned inwardly from outer wall **807** of capsule housing **805**. In FIG. **13**B, payload coupling apparatus **800**′ is in an extended state where end **806***a*′ of lip or hook **806**′ has been moved outwardly from capsule housing **805** such that the end **806***a* of the lip or hook **806**′ is positioned outwardly from outer wall **807** of capsule housing **805**. Lip of hook **806**′ may be moved outwardly via cams or protrusions within channel **1050**, or by a spring-loaded portion of channel **1050**, or other mechanisms. In the extended state shown in FIG. **13**B, the hook or lip **806**′ is in position to easily extend through the aperture **513** in handle **511** of payload **510**, such that the handle **511** is positioned within slot **808** of payload coupling apparatus **800**′ and retrieval of the payload and removal from the payload holder of the payload retrieval apparatus can be achieved. Once the payload **510** is removed from the payload holder the hook or lip **806**′ may be moved back to its retracted state as shown in FIG. **13**A.

(215) FIG. **13**C is a side view of payload coupling apparatus **800**″ which in this illustrative embodiment is the similar to payload coupling apparatus **800** shown in FIGS. **6**A-**6**C, but instead includes a plurality of magnets **830** positioned thereon. The plurality of magnets **830** are adapted to magnetically engage a plurality of magnets **1060** (or a metal) positioned within the channel **1050** of a payload retrieval apparatus **1000** as shown in FIG. **20** below to orient the payload coupling apparatus **800**″ within the channel **1050** of payload retrieval apparatus **1000** so that the hook or lip **806***a* is in proper position to extend through aperture **513** of handle **511** of payload **510** to effect removal of payload **510** from the payload holder of payload retrieval apparatus **1000**.

(216) FIG. **13**D is a side view of payload coupling apparatus **900** which in this illustrative embodiment is similar to payload coupling apparatus **800**″ shown in FIG. **6**C, but instead includes a weighted side **840**. The weighted side **840** serves to orient the payload coupling apparatus **900**

within the channel **1050** of payload retrieval apparatus **1000** so that the hook or lip **806***a* is in proper position to extend through aperture **513** of handle **511** of payload **510** to effect removal of payload **510** from the payload holder of payload retrieval apparatus **1000**.

(217) In each of the payload coupling apparatuses **800**, **800′**, **800″**, and **900** described above, the upper and lower ends are rounded, or hemispherically shaped, to prevent the payload coupling apparatus from snagging during descent from, or retrieval to, the fuselage of a UAV. Furthermore, each of payload coupling apparatuses **800**, **800″**, and **900** may have a retractable and extendable hook or lip as is shown in FIGS. **13**A and **13**B with regard to payload coupling apparatus **800′**.

(218) In addition, as illustrated in FIG. **9**, the payload delivery system may automatically align the top portion of the payload during winch up, orienting it for minimum drag along the aircraft's longitudinal axis. This alignment enables high speed forward flight after pick up. The alignment is accomplished through the shape of the payload hook and receptacle. In the payload coupling apparatus **800**, the lower lip or hook **806** has cam features around its perimeter which always orient it in a defined direction when it engages into the cam features inside the receptacle of the fuselage of the UAV. The tips of the cam shapes on both sides of the capsule are asymmetric to prevent jamming in the 90 degree orientation. In this regard, helical cam surfaces may meet at an apex on one side of the payload coupling mechanism, and helical cam surfaces may meet at a rounded apex on the other side of the payload coupling mechanism. The hook is specifically designed so that the package hangs in the centerline of the hook, enabling alignment in both directions from 90 degrees.

(219) Payload coupling apparatuses **800**, **800′**, **800″**, and **900** include a hook **806** (or **806′**) formed beneath a slot **808** such that the hook also releases the payload passively and automatically when the payload touches the ground upon delivery. This is accomplished through the shape and angle of the hook slot and the corresponding handle on the payload. The hook slides off the handle easily when the payload touches down due to the mass of the capsule and also the inertia wanting to continue moving the capsule downward past the payload. The end of the hook is designed to be recessed slightly from the body of the capsule, which prevents the hook from accidentally re-attaching to the handle. After successful release, the hook gets winched back up into the aircraft.

(220) FIGS. **14-16** are perspective views of payload retrieval apparatus **1000** having a payload **510** positioned thereon, according to an example embodiment. The payload retrieval apparatus **1000** may be a non-permanent structure placed at a payload retrieval site. The apparatus includes an extending member **1010** that may be secured to a base or stand **1012** at a lower end of the extending member **1010**. Alternately, the extending member **1010** may have a lower end that may be positioned within a corresponding hole in the ground or hole in an apparatus positioned on the ground. The payload retrieval apparatus **1000** may be readily folded up, like an umbrella stand, to provide for ease of transport. In addition, because of its non-permanent configuration, payload retrieval apparatus **1000** may not require any type permitting, which may not be the case for a permanent device used for UAV loading and unloading.

(221) An angled extender **1020** may be attached at an upper end of the extending member **1010**, and adapter **1016** may be used to adjust the height or angle of the angled extender **1020**, and having a threaded set screw with knob **1018** to set the angled extender **1020** into a desired position. The angled extender **1020** is shown with an upper end secured to a channel **1050**. A first end of the channel may have a first extension or tether engager **1040** that extends in a first direction from a lower end of the channel **1050** and a second extension or tether engager **1030** that extends in a second direction from the lower end of the channel **1050**. A second end of the channel **1050** may have a payload holder **570**, **572** positioned near or thereon that is adapted to secure a payload **510** to the second end of the channel **1050**.

(222) A shield **1042** is shown extending from the first tether engager **1040**, and another shield **1032** is shown extending from the second tether engager **1030**. Shield **1042** and **1032** may be made of a fabric material, or other material such as rubber or plastic. A shield **1052** is also shown extending from the first end of channel **1050**. Shields **1042**, **1032**, and **1052** serve to prevent a payload

retriever **800** extending from an end of a tether **1200** attached to a UAV from wrapping around the tether engagers **1040** and **1032** or other components of payload retrieval apparatus **1000** when the payload retriever comes into contact with tether engagers **1040** or **1030** during a payload retrieval operation.

(223) Channel **1050** includes a tether slot **1054** extending from a first end to a second end of the channel **1050**, and the tether slot **1054** allows for a payload retriever to be positioned within the channel **1050** attached to a tether which extends through the tether slot **1054**. A payload holder is shown that is a pair of pins **570**, **572** that extend through openings in handle **511** of payload **510** to suspend payload **510** in position adjacent the second end of the channel **1050** ready to be retrieved by a payload retriever attached to a tether suspended from a UAV.

(224) To provide for automatic retrieval of payload **510** with a payload retriever suspended from a UAV with a tether, payload **510** is secured to the payload holder **570**, **572** on the second end of the channel **1050** at the payload retrieval site. A UAV arrives at the payload retrieval site with a tether **1200** extending downwardly from the UAV and with the payload retriever **800** positioned on the end of the tether, as shown in FIGS. **14** and **17**. The UAV approaches the payload retrieval apparatus **1000**, and as it nears the payload retrieval apparatus **1000**, the tether **1200** comes into contact with the first or second extension (tether engager) **1040**, **1030**. As the UAV moves forward, or the UAV is moved upwardly, or the payload retriever is winched upwardly to the UAV while the UAV is hovering in place (or any combination thereof), the tether slides inwardly along the first or second extension **1040**, **1030** where it is directed towards the first end of the channel **1050**. With further forward or upward movement of the UAV, or upward winching of the payload retriever, the tether **1200** moves through the tether slot **1054** of channel **1050** and eventually the payload retriever **800** attached to the tether **1200** is pulled into the channel **1050** by the tether. The payload retriever **800** is pulled through the channel **1050** where it engages, and secures, the payload **510** secured to the payload holder **570**, **572**. The payload retriever **800** then pulls the payload **510** free from the payload holder **570**, **572**. Once the payload **510** is free from the payload holder **570**, **572**, the payload **510** may be winched upwardly into secure engagement with the UAV, and the UAV may continue on to a delivery site where the payload **510** may be delivered by the UAV.

(225) FIG. **17** shows a sequence of steps A-D performed in the retrieval of payload **510** from payload retrieval apparatus **1000**, shown in FIGS. **14-16**. A payload retriever, shown in FIG. **17** as payload coupling apparatus **800** having a hook or lip **806** positioned beneath slot **808**, is attached to an end of tether **1200** which is in turn attached to a UAV. At point A in the sequence of steps shown from right to left, payload retriever **800** is shown suspended at the end of tether **1200** at a position below the height of tether engagers **1040** and **1030**. Payload retriever **800** and tether **1200** move towards the payload retrieval apparatus **1000**, where tether **1200** contacts tether engager **1040** or tether engager **1030**, and tether **1200** and payload retriever **800** move towards channel **1050** until payload retriever **800** is positioned just outside of channel **1050** shown at point B in the sequence. With further forward or upward movement of the UAV, or upward winching of payload retriever **800** (or any combination thereof), tether **1200** extends through tether slot **1054** of channel **1050** and payload retriever **800** is positioned within channel **1050** as shown at point C of the sequence. With further forward or upward movement of the UAV, or upward winching of the payload retriever **800** (or any combination thereof), payload retriever **800** exits channel **1050** and hook or lip **806** of payload retriever **800** engages handle **511** of payload **510** and removes payload **510** from payload holder **570**, **572** positioned on the end of the channel **1050**. After removal of payload **510** from payload holder **570**, **572** of payload retrieval apparatus **1000**, at point D of the sequence, payload **510** is suspended from tether **1200** with handle **511** of payload **510** positioned in slot **808** above hook or lip **806** of payload retriever **800**, where payload **510** may be winched up to the UAV and flown for subsequent delivery at a payload delivery site.

(226) FIG. **18** is a perspective view of payload retrieval apparatus **1000** shown in FIGS. **14-17** with a payload loading apparatus **1080** having a plurality of payloads **510-2** and **510-3** positioned

thereon, according to an example embodiment. Payload loading apparatus **1080** includes a platform **1082** positioned on platform base **1086** having an upper surface **1084** that downwardly slopes towards payload retrieval apparatus **1000**. Payload loading apparatus **1080** allows for automatic loading of a subsequent payload positioned on upper surface **1084** of payload loading apparatus **1080** onto payload retrieval apparatus **1000** after a payload positioned on the payload holder has been retrieved. In particular, once payload **510-1** has been removed from payload holder **570**, **572** of payload retrieval apparatus **1000**, subsequent payload **510-2** slides down the upper surface **1084** of the payload loading apparatus **1080** and is secured to payload holder **570**, **572** of payload retrieval apparatus **1000**. Payload loading apparatus **1080** may include one or more rollers **1088** that provide for the downward movement of upper surface **1084**, like a conveyor belt.
(227) As shown in FIG. **18**, the handle **511** of payload **510-1** has openings **524** and **526** (see FIG. **11**A) through which pins **570**, **572** extend to hold payload **510-1** in position for retrieval. However, handle **511** may also include magnets **524′** and **526′** (see FIG. **11**B) that are adapted to magnetically engage corresponding magnets or a metal positioned on the payload holder of the payload retrieval apparatus **1000**. With a magnetic handle, the magnets **524′** and **526′** on the handle **511** move into engagement with the payload holder to hold subsequent payload **510-2** into position for subsequent retrieval as illustrated in the sequence of steps at points A-D shown in FIG. **17**. In addition, payloads **510-1** through **510-3** may include fiducials **585** that may take the form of an RFID tag or bar code to identify the contents of the payload and delivery site information and/or delivery instructions. As a result, using payload loading apparatus **1080** in conjunction with payload retrieval apparatus **1000**, a plurality of payloads may be retrieved from payload apparatus **1000** without the need for a person to reload subsequent payloads for retrieval, providing for further automated payload retrieval.
(228) In order for the hook or lip **806** of the payload retriever **800** (shown in FIGS. **6**A-C) to engage the handle **511** of payload **510** to effect removal and retrieval of the payload **510** from the payload retrieval apparatus **1000**, the hook or lip **806** should be positioned downwardly when it exits the channel **1050** in the embodiment shown (different orientations are possible in alternate embodiments). As illustrated in FIG. **19**, to ensure that the slot hook or lip **806** of the payload retriever **800** is in a proper orientation as the payload retriever **800** exits the channel **1050** and engages the handle **511** of the payload **510**, the payload retriever **800** may be provided with exterior cams **804** or slots that correspond to cams or slots **1058**, **1059** positioned on an interior surface of the channel **1050**. The interaction of the cams **804** or slots on the payload retriever **800** and cams or slots **1058**, **1059** on the interior of the channel **1050** properly orient the payload retriever **800** within the channel **1050** such that hook or lip **806** beneath the slot **808** of the payload retriever **800** is in proper position to extend through the aperture **513** on the handle **511** of the payload **510** to remove the payload **510** from the payload holder **570**, **572**.
(229) FIG. **19** is a perspective view of channel **1050** of the payload retrieval apparatus **1000** shown in FIGS. **14-16** with a payload retriever **800** positioned therein. Channel **1050** includes a tether slot **1054** through which tether **1200** extends when tether **1200** draws payload retriever **800** into the interior of channel **1050**. The interior of channel **1050** includes cams or slots **1058**, **1059** which cooperate with cams **804** or slots on the payload retriever **800** to properly orient the hook or lip **806** and slot **808** in a downward facing position within the channel **1050**. Thus, the interaction of cams or slots **1058**, **1059** on the interior of channel **1050** with cams **804** or slots on the payload retriever **800** provides a desired orientation of the payload retriever **800** at the point that payload retriever **800** exits the channel **1050** and engages handle **511** of payload **510** to remove the payload **510** from the payload holder **570**, **572**.
(230) Alternately, or in addition to cams **804**, the payload retriever **800″** may have one or more magnets **830** positioned thereon as shown in FIGS. **13**C and **20** that cooperate with one or more magnets **1060**, or a metal, positioned on an interior of the channel **1050** and magnetic interaction is used to properly orient the payload retriever **800″** within the channel **1050** during the process of

payload retrieval.

(231) FIG. **20** is a perspective view of channel **1050** of the payload retrieval apparatus **1000** shown in FIGS. **14-16** with a payload retriever **800**″ positioned therein. Channel **1050** includes a tether slot **1054** through which tether **1200** extends when tether **1200** draws payload retriever **800**″ into the interior **1056** of channel **1050**. The interior **1056** of channel **1050** includes a plurality of magnets **1060** which magnetically engage with magnets **830**, or a metal, on the payload retriever **800**″ to properly orient the hook or lip **806** and slot **808** in a downward facing position within the channel **1050**. Thus, the interaction of magnets **1060** on the interior **1056** of channel **1050** with magnets **830** or simply a metal on the payload retriever **800**″ provides a desired orientation of the payload retriever **800**″ at the point that payload retriever **800**″ exits the channel **1050** and engages handle **511** of payload **510** to remove the payload **510** from the payload holder **570**, **572**. Alternatively, or in addition, a metal strip or plurality of metal pieces could be positioned within the channel **1050** to provide for magnetic engagement with the magnets **830** on the payload retriever **800**″. Similarly, one or more magnets may be positioned on the interior of channel **1050** that magnetically engage a metal positioned on a payload retriever.

(232) In addition, the payload retriever could be weighted to have an offset center of gravity (see payload retriever **900** shown in FIG. **13**D) such that the hook **806** and slot **808** of the payload retriever **900** are positioned properly (with the “heavy” portion of the capsule on a lower side) to engage the handle **511** of the payload **510** and effect removal of the payload **510** from the payload holder **570**, **572**. The weighted side **840** of payload retriever **900** helps to insure that the hook or lip **806** and slot **808** are positioned downwardly within the channel **1050** so as to be in position for the hook or lip **806** to extend through aperture **513** in handle **511** of payload **510** during the retrieval process. It will be appreciated that the use of cams, magnets, and a weighted side could all be used separately, or used in combination in whole or in part, to provide for a desired orientation of the payload retriever within the channel to effect removal of the payload from the payload retrieval apparatus **1000**.

(233) As shown in FIG. **21**A, the channel **1050** may also have an interior that tapers downwardly, or decreases in size, as the channel **1050** extends from the first end where the payload retriever enters the interior **1056** of channel **1050** to the second end where the payload retriever exits the channel **1050** to further facilitate the proper orientation of the payload retriever within the channel. In addition, as shown in FIG. **21**B, the second end of the channel **1050** could be spring loaded with a spring **1061** exerting a force against outer surface **1057** of channel **1050**, or operate as a leaf spring, to also facilitate the proper orientation of the payload retriever (or extension or the hook or lip of the payload retriever) at the point of payload retrieval.

(234) Not only does the payload retrieval apparatus **1000** described above provide for automatic payload retrieval without the need for human involvement, but the UAV advantageously is not required to land for the payload **510** to be loaded onto the UAV at the payload retrieval site. Thus, the UAV may simply fly into position near the payload retrieval apparatus **1000** and maneuver itself to position the tether **1200** between the first and second tether engagers **1040**, **1030**, which may be aided by the use of fiducials (which could take the form of an RFID tag or bar code) positioned on or near the payload retrieval apparatus **1000** and/or an onboard camera system positioned on the UAV. Once in position, the UAV may then move forward or upward, or the payload retriever may be winched up towards the UAV (or any combination thereof) to pull the payload retriever through the channel **1050** and into engagement with the handle **511** of the payload **510** and effect removal of the payload **510**. In some payload retrieval sites, landing the UAV may be difficult or impractical, and also may engage with objects or personnel when landing. Accordingly, allowing for payload retrieval without requiring the UAV to land provides significant advantages over conventional payload retrieval methods.

(235) FIG. **22** is a side view of payload retrieval apparatus **1400**, which includes a base **1402** and an upwardly extending member **1404**. Also included is a first sloped surface **1410** and a second

sloped surface **1420**. A first channel **1433** is defined between first sloped surface **1410** and surface **1430** and is positioned above upwardly extending member **1404**. An opening **1432** is provided into first channel **1433**. A payload **510** is positioned at an end of first channel **1433**. A second channel **1424** is provided having a wall **1422** extending downwardly from second sloped surface **1420**. First and second sloped surfaces **1410** and **1420** provide a funneling system for a payload retrieval **800** attached to a tether **1200** and serves to funnel payload retrieval **800** towards opening **1432** in first channel **1433**.

(236) FIG. **23** is a top view of payload retrieval apparatus **1400**. Sloped surfaces **1460** and **1462** are provided with a tether slot **1450** positioned therebetween. Opening **1432** to channel **1433** is shown with payload **510** positioned beneath sloped surfaces **1460** and **1462**.

(237) FIGS. **24**A-E illustrate a sequence of steps used to automatically pick up payload **510**. In FIG. **24**A, payload retriever **800** attached to tether **1200** is shown descending towards the funneling system formed by first sloped surface **1410** and second sloped surface **1420**. FIG. **24**B illustrates payload retriever **800** landing on first sloped surface **1410**. The payload retriever will then slide down first sloped surface towards opening **1425** between first sloped surface **1410** and second sloped surface **1420**. FIG. **24**C illustrates payload retriever **800** after it has slid down first sloped surface **1410**, through opening **1425** and into second channel **1424**. While positioned in second channel **1424**, payload retriever **800** is positioned for entry through opening **1432** into first channel **1433**. In FIG. **24**D, payload retriever **800** has been winched upwardly into first channel **1433**, where it is positioned to move further upwardly to secure handle **511** of payload **510**. In FIG. **24**E, payload retriever **800** has moved further upwardly to secure payload retriever **800** to handle **511** of payload **510**, where payload **510** can be removed from the end of the first channel **1433** and winched up to a UAV for transport.

(238) FIG. **25**A is a perspective view of payload retrieval apparatus **1480**. Payload retrieval apparatus **1480** includes a base **1402** with a cross member **1406** and truss members **1407** and **1408**. Upwardly extending member **1404** is attached to base **1402**. A first sloped surface **1460** is positioned adjacent to second sloped surface **1462** are attached to member **1465** (attached to upwardly extending member **1404**) with a tether slot **1450** positioned therebetween. Opening **1470** extends towards a channel positioned on member **1465**, or beneath the first and second sloped surfaces **1460** and **1462**, which is adapted to receive payload retriever **800**. First and second sloped surfaces **1460** and **1462** serve as a funneling system to funnel a payload retriever **800** downwardly towards opening **1470**, where a tether **1200** may move the payload retriever **800** into position to extend through opening **1470** into a channel positioned on member **1465**, or beneath the first and second sloped surfaces **1460** and **1462**, and tether **1200** extends through the tether slot **1450** to draw the payload retriever **800** towards a payload for automated payload retrieval. The payload retriever **800** may land anywhere on either of the first or second sloped surfaces **1460**, **1462**, and will funnel down until it slides off of the sloped surfaces, where the tether **1200** may be drawn through tether slot **1450** to draw the payload retriever **800** into engagement with the handle **511** of payload **510**. First and second sloped surfaces **1460** and **1462** provide a V-shaped funneling system that is downwardly sloped towards opening **1470**. It will be appreciated that the sloped surfaces in payload retrieval apparatus **1400** and **1480** may have other configurations and geometries to provide a funneling system for the payload retriever **800**. The surfaces may be hard or soft, or even made of netting to reduce wind load. Furthermore, the surfaces are not required to be flat, but could be rounded or concave as well.

(239) In addition, the first and second sloped surfaces **1460** and **1462** are downwardly sloped towards opening **1470** to a channel. The bottoms of the first and second sloped surfaces are also positioned at an angle towards opening **1470**. In applications where the payload retriever does not land on either of sloped surfaces **1460** or **1462**, the tether **1200** descend in front of opening **1470** and may be drawn towards opening **1470** along the angled lower surfaces of the first and second sloped surfaces **1460** and **1462**. The tether **1200** may be drawn, or simply slide, down the angled

lower surfaces until the tether **1200** is in front of the tether slot **1450**. At this point, the tether **1200** may be drawn through the tether slot **1450**, thereby drawing the payload retriever **800** into the channel. It should also be noted that first and second sloped surfaces **1460** and **1462** not only serve to provide a funneling system to funnel the payload retriever **800** towards opening **1470**, but also serve to block wind from blowing the payload retriever **800** out of position.
(240) FIG. **25**B is a side view of payload retriever apparatus **1480**, and includes cross member **1406**, truss **1407**, and upwardly extending member **1404**. Member **1465** extends from upwardly extending member **1404** with second sloped surface **1462** positioned thereon. A channel with a curved portion **1439** is positioned on an end of member **1465**, with a payload **510** positioned on curved portion **1439** of the channel. Although the channel is positioned beyond second sloped surface **1462**, the channel could also extend beneath second sloped surface **1462**.
(241) FIG. **25**C is a side view of an end of the payload retrieval apparatus **1480**. A channel is shown extending from member **1465** with a payload **510** positioned on curved portion **1439**.
(242) FIG. **25**D is a perspective view of an end of the payload retrieval apparatus **1480**. A channel with a curved portion **1439** is positioned on an end of member **1465**, with a payload **510** positioned on curved portion **1439** of the channel. Handle **511** of payload **510** is positioned on pins **570** and **572** extending from curved portion **1439** of the channel.
(243) FIG. **25**E shows perspective views of payload retrieval apparatus **1500**. Payload retrieval apparatus **1500** includes a base **1510** and upwardly extending side walls **1520**. A first sloped surface **1560** is positioned adjacent second sloped surface **1562** are positioned within side walls **1520** with a tether slot **1550** positioned therebetween. Opening **1570** extends towards a channel positioned beneath or near the first and second sloped surfaces **1560** and **1562** which is adapted to receive payload retriever **800**. First and second sloped surfaces **1560** and **1562** serve as a funneling system to funnel a payload retriever **800** downwardly towards opening **1570**, where a tether **1200** may move the payload retriever **800** into position to extend through opening **1570** into a channel beneath or near the first and second sloped surfaces **1560** and **1562**, and tether **1200** extends through the tether slot **1550** to draw the payload retriever **800** towards a payload for automated payload retrieval.
(244) FIG. **26** is a perspective view of payload retrieval apparatus **1480**. Member **1465** is rotatable with respect to upwardly extending member **1404** to allow the first and second sloped panels **1460** and **1462** to rotate with the wind such that the stand **1480** is positioned into the wind to reduce the impact of wind on payload retrieval apparatus **1480**. As with a non-rotatable payload retrieval apparatus, first and second sloped surfaces **1460** and **1462** not only serve to provide a funneling system to funnel the payload retriever **800** towards opening **1470**, but also serve to block wind from blowing the payload retriever **800** out of position.
(245) FIG. **27**A shows perspective views of rotational spring loaded pusher **1600**. Rotational spring loaded push **1600** is positioned near the end of channel **1433** formed between edges **1410** and **1430**, with edge **1430** having a shorter length than edge **1410**. As the payload retriever **800** exits the channel **1433**, the spring loaded pusher **1600**, rotatable about pivot point **1620**, includes a cam **1610** that initially comes into contact with a top surface of payload retriever **800**. As payload retriever **800** exits channel **1433**, the spring loaded cam **1610** pushes against a bottom of the payload retriever **800**, to force lip **806** of payload retriever **800** forward into engagement with handle **511** of payload **510**.
(246) FIG. **27**B shows a side view of leaf spring **1640**. Leaf spring **1640** operates in a similar manner to rotational spring loaded pusher **1600**. As payload retriever **800** exits channel **1433**, the leaf spring **1640** pushes against a bottom of the payload retriever **800**, to force lip **806** of payload retriever **800** forward into engagement with handle **511** of payload **510**. The leaf spring **1640** may be a separate metal spring, or molded-in plastic tabs that deform to impart a spring force on the payload retriever **800**.
(247) FIG. **27**C shows a side view of linear spring plunger **1650**. Linear plunger **1650** includes

spring **1654** and protrusion **1652**. Linear spring plunger **1650** operates in a similar manner to rotational spring loaded pusher **1600** and leaf spring **1640**. As payload retriever **800** exits channel **1433**, the protrusion **1652** of linear spring plunger **1650** pushes against a bottom of the payload retriever **800**, to force lip **806** of payload retriever **800** forward into engagement with handle **511** of payload **510**.
(248) FIG. **28** is a perspective view of payload retrieval apparatus **1700**. Payload retrieval apparatus **1700** provides a bowl-shaped funneling system **1720**. A payload retriever **800** descends onto the funneling system **1720** and slides down through lower opening **1760**. The tether **1200** attached to the payload retriever is drawn through tether slot **1750** until payload retriever connects with handle **511** of payload **510** to secure the payload **510** to payload retriever **800** for removal of payload **510** from the payload retrieval apparatus **1700**. Advantageously, payload retrieval apparatus **1700** may accommodate multiple payloads **510**. As shown in FIG. **28**, one payload **510** is positioned in a northern position and another payload **510** is shown in a southern position. A second tether slot may be provided for access to the southern payload **510** such that the payload retriever **800** may travel beneath tether slot **1750** to the northern payload **510**, or beneath the second tether slot to pick up the southern payload **510**. Additional payloads could also be provided on payload retrieval apparatus **1700**. For example, eastern and western payloads could be included with corresponding eastern and western tether slots.
(249) FIGS. **29**A-B show perspective and side views of spring loaded plunger pin **1484**. Spring loaded plunger **1484** extends into channel **1433**, along with an oppositely disposed plunger pin (not shown). As a payload retriever **800** comes into contact with plunger pin **1484**, the payload retriever **800** is rotated into a desired position such that the lip **806** of the payload retriever is properly positioned to engage with an opening in the handle **511** of the payload upon exiting the channel **1433**.
(250) FIGS. **30**A-B show side views of protrusions **1519**. Protrusions **1519** operate in a similar manner to rotational spring loaded pusher **1600**, leaf spring **1640**, and linear spring plunger **1650** shown in FIGS. **27**A-C. As payload retriever **800** exits channel **1433**, the protrusions **1519** push against a bottom of the payload retriever **800**, to force lip **806** of payload retriever **800** forward into engagement with handle **511** of payload **510**.
(251) FIGS. **31**A-B show side and perspective views of curved portion **1439**, FIGS. **32**A-C show side and perspective view of curved portion **1439**, and FIGS. **33**A-B show perspective views of curved portion **1439**. Channel **1433** between edges **1410** and **1430** ends with a curved portion **1439**. Payload retriever **800** initially travels through channel **1433** along a centerline of the channel. However, curved portion **1439** changes the angle of exit of payload retriever **800** from channel **1433**. Curved portion **1439** provides significant advantages over an entirely straight channel. The curved portion **1439** at the end of the channel **1433** angles the payload retriever **800** upon exiting the channel **1433** to have the payload retriever **800** “lean back” such that the lip **806** of the payload retriever **800** extends towards the opening **513** in the handle **511** of the payload **510**. The curved portion **1439** also allows for a top of the payload retriever **800** to contact the handle **511** such that a portion of the handle **511** over the opening **513** in the handle **511** contacts the payload retriever **800** and the portion over the opening **513** slides down the payload retriever **800** until the lip **806** of the payload retriever **800** extends into the opening **513** in the handle **511** of the payload **510**.
(252) In FIG. **32**B, payload holder in the form of extending pins **570** and **572** is shown. In FIG. **32**C, handle **511** of the payload **510** is shown positioned on extending pins **570** and **572**. Lip **806** of payload retriever **800** is shown extending through opening **513** in handle **511**. The handle **511** of the payload **510** itself may act as a spring upon entry of the lip **806** into the opening of the handle **511** of the payload **510** to rotate the payload retriever **800** into the proper position. For example, if the rotational position of the payload retriever **800** is off somewhat, then the handle **511** of the payload **510** itself may act to rotate the payload retriever **800** into its desired rotational position. FIGS. **33**A and **33**B illustrate that the angle of the channel may be altered, for example, between 45

and 60 degrees. The change in angle of the channel can also provide for the positioning of the lip **806** of the payload retriever **800** to be in an improved position for the lip **806** to extend into an opening **513** in the handle **511** of a payload **510**. In particular, when the channel is at a 60 degree angle, the lip **806** of the payload retriever **800** extends further outwardly to extend through the opening in handle **511** of the payload **510**.

(253) FIGS. **34**A-E show various perspective views of pivoting carriage **1800**. Pivoting carriage **1800** includes payload retrieval holder **1802** that pivots about pivot **1804**. Pivoting carriage **1800** uses payload retrieval holder **1802** to hold payload retriever **800**. Payload retrieval holder **1802** pivots downwardly about pivot **1804** to place lip **806** of payload retriever **800** through opening **513** in handle **511** of payload **510**. After the payload retriever **800** is secured to handle **511** of payload **510**, the payload retriever **800** may be removed from the payload retriever holder **1802** to remove payload **510** from its position.

(254) FIG. **35** illustrates that a UAV positioned at 7 meters above the ground may be used to allow a payload retriever **800** to remove payload **510** from a payload retrieval apparatus such as payload retrieval apparatus **1000** shown in FIG. **35**.

(255) FIGS. **36-40**D shows various views of payload retrieval apparatus **1900**. Payload retrieval apparatus **1900** is used for automated payload pickup using a UAV. A payload **1935** is positioned on a payload holder on a rear end of payload retrieval apparatus **1900**. Payload retrieval apparatus **1900** includes base **1904**, upwardly extending member **1910**, and a payload coupling apparatus channel **1940** housed within enclosure **1920**. Payload retrieval apparatus **1900** also includes tether engagers **1930** and **1932** which are used to engage a tether attached to a payload coupling apparatus, whereafter the payload coupling apparatus is drawn into and through the payload coupling apparatus channel **1940** to pick up payload **1935**. Tether engager **1930** includes member **1933** which provides mechanical support for tether engager **1930**, and provides other functions. In the same manner, tether engager **1932** includes member **1934** which provides mechanical support for tether engager **1932**, and provides other functions. Payload retrieval apparatus **1900** provides for automated pickup of payload **1935**, and may operate in the same or a similar manner as payload retrieval apparatuses **1000** and **1480** described above.

(256) Tether engager **1930** includes an upper guide member **1933** that is configured to help maintain the end of the tether in a substantially vertical orientation as the payload coupling apparatus is drawn through the payload coupling apparatus channel **1940**. With the inclusion of upper guide member **1933**, tether engager **1930** includes both an upper edge and a lower edge for guiding the tether as the payload coupling apparatus is received and drawn through the payload coupling apparatus channel **1940**. The lower edge is formed by the primary member of tether engager **1930** and extends toward a receiving end of payload coupling apparatus channel **1940**. Accordingly, the lower edge of tether engager **1930** directs a portion of the tether that is near the payload coupling apparatus to the receiving end of the channel **1940**.

(257) The upper guide member **1933**, on the other hand, extends towards payload coupling apparatus channel **1940** at an elevated height compared to the lower edge formed by the primary member of tether engager **1930**. This allows the upper guide member **1933** to engage a portion of the tether that is spaced from the payload coupling apparatus at a position that is elevated above the channel **1940**. Accordingly, when the payload coupling apparatus is within the channel **1940**, the direction of the portion of the tether that extends upward from the channel **1940** will be substantially vertical. Thus, even if the UAV is laterally offset from the position of the payload retrieval apparatus **1900**, such that most of the length of the tether extending between the UAV and the payload retrieval apparatus **1900** is at substantial angle, the end portion of the tether that extends from the channel **1940** will maintain a substantially vertical orientation. With this end portion of the tether in a substantially vertical orientation, the tension on the tether as it is retracted can effectively pull the payload coupling apparatus through the payload coupling apparatus channel **1940**.

(258) Similar to tether engager **1930**, tether engager **1932** also includes an upper guide member **1934** with a similar configuration that is operable to maintain an end portion of the tether in a substantially vertical orientation.
(259) In addition to helping maintain the orientation of the tether, the upper guide members **1933**, **1934** may also provide structural support to the respective tether engagers. For example, because of the inclusion of upper guide member **1933** in tether engager **1930**, the tether engager **1930** is secured to upwardly extending member **1910** at two independent points. The primary member of tether engager **1930** is secured to the upwardly extending member **1910** at a lower position and the upper guide member **1932** is secured to the upwardly extending member **1910** at an upper position. Furthermore, a triangular frame is formed between the upwardly extending member **1910**, the primary member of tether engager **1930**, and the upper guide member **1932**, which provides a strong support structure for tether engager **1930**.
(260) While tether engager **1930**, as shown in FIGS. **36-40**D, is formed as a frame, such that upper guide member **1933** is formed as a second pole or rod that extends at an angle from the primary member (or pole) of tether engager **1930**, other configurations are possible. For example, in some embodiments, the tether engager may be formed by a similarly shaped structure with a continuous surface for guiding the tether. In such an embodiment, a lower edge of the tether engager may provide a lower guide, while an upper edge of the tether engager may form the upper guide member that maintains the substantially vertical orientation of the tether, as described above.
(261) FIG. **39** illustrates that payload retrieval apparatus **1900** may advantageously be sized to span only a single parking space **1937**.
(262) FIGS. **41**A-C show various views of payload retrieval apparatus **1950**. Payload retrieval apparatus **1950** operates in a manner similar to payload retrieval apparatus **1900**. Payload retrieval apparatus **1950** includes base **1954**, upwardly extending member **1960**, and a payload coupling apparatus channel **1990**. Payload retrieval apparatus **1950** also includes tether engagers **1980** and **1982** which are used to engage a tether attached to a payload coupling apparatus, whereafter the payload coupling apparatus is drawn into and through the payload coupling apparatus channel **1990** to pick up payload **1985**. Payload coupling apparatus channel **1990** may include a guiding member with a tether slot which the payload coupling apparatus rides beneath until it is drawn into a curved portion of the payload coupling apparatus channel **1990** attached to the guiding member. Payload retrieval apparatus **1950** is shown with a shield **1992** which helps to prevent the payload coupling apparatus from getting tangled with the frame of the payload retrieval apparatus **1950**. Payload retrieval apparatus **1950** provides for automated pickup of payload **1985**, and may operate in the same manner as payload retrieval apparatuses **1000** and **1480** described above.
(263) FIG. **42** is a close-up rear view of payload retrieval apparatus **1900**, and FIG. **43** is a close-up perspective rear view of payload retrieval apparatus **1900**. A channel passage **1962** extends within channel **1960** and extends through to the front of payload retrieval apparatus **1900** as shown in FIG. **44**. Channel **1960** includes a tether slot **1964** to allow for the passage of a tether attached to a payload retriever as the payload retriever passes through the channel **1960**. On the rear side of payload retrieval apparatus **1900**, channel **1960** has a funnel shape which narrows from an entry point as it extends towards the front of payload retrieval apparatus **1900**.
(264) FIG. **44** is a close-up perspective front view of payload retrieval apparatus **1900**. A pair of pins **1927** and **1929** are positioned on payload holder wall **1925** at the exit of channel passage **1962**, that are configured to extend through corresponding apertures in the handle of a payload as described in more detail above. Therefore, the pair of pins **1927** and **1929** serve as a payload holder to hold a payload that is in position to be retrieved by a payload retriever as the payload retriever exits channel portion **1970** through channel passage **1962**. Payload retrieval apparatus **1900** includes side plates **1920** and top plate **1922** that serve to protect the channel **1960** from wind, debris, and other elements that could interfere with the proper operation of the channel. Top plate **1922** may also serve to guide a tether attached to a payload retriever through tether slot **1964** in

channel **1960**.

(265) When a payload is positioned on a payload holder of a payload retrieval apparatus, there is a concern for the security of the payload. For example, an unattended payload could be subject to theft or tampering. Therefore, it would be desirable to provide security measures regarding the payload to prevent theft or tampering of the payload **2150** or its contents, to ensure that the payload delivery is completed as planned. One option is to secure the entire payload retrieval apparatus within a fenced area behind a locked gate to secure the payload to be retrieved. Alternatively, there are physical measures that can be taken on the payload retrieval apparatus itself that may be employed to secure the payload that are described below.

(266) FIG. **45**A is a perspective view of payload retriever **800** passing through channel passage **1962** of channel **1960** towards locking member **2040**. Locking member **2040** is mounted on pivot **2042**. A first end **2044** of locking member **2040** engages a latch **2052** on locking cover **2055** to lock locking strap **2055** into a locked position over handle **511** of payload **2150** and pins **570** and **572** of payload holder **2060**. When positioned in the locked position shown in FIG. **45**A, locking cover **2055** encloses the handle **511** and serves to prevent the theft of payload **2150**, as well as prevent the opening of payload **2150** and any tampering with the contents within payload **2150**.

(267) FIG. **45**B is a perspective view of a locking member **2040** having a second movable end **2046** extending through a wall of channel **1960** into an interior of channel **1960**. Payload retriever **800** attached to tether **2162** is shown passing through channel passage **1962** of channel **1960** and over the movable second end **2046** of locking member **2040** to unlatch the first end **2044** of locking member **2040** from latch **2052** and move locking member **2040** to an unlocked position where locking cover **2055** is moved away from the payload handle **511** of payload **2150**. As illustrated in FIG. **45**B, in the unlocked position, the movable second end **2046** of locking member **2040** is depressed by payload retriever **800** as payload retriever **800** passes through the channel **1960** to unlatch the first end **2044** from latch **2052** on locking cover **2055**. The locking member **2040** may be spring-biased such that the second movable end **2046** is biased towards the interior of channel **1960**. In addition, locking cover **2055** may be spring-biased into the open position shown in FIG. **45**B. The locking member **2040** thus acts as a switch to release the locking cover **2055** from the locked position as the payload retriever **800** passes through the channel passage **1962**.

(268) While the latch **2052** that holds the locking cover **2055** shut is illustrated as being on top of locking cover **2055**, in some embodiments the latch may be positioned inside the locking cover **2055** so that the locking member **2040** is covered and cannot be unlatched from the outside.

(269) FIG. **45**C is a perspective view of payload retriever **800** exiting the channel passage **1962** of channel **1960** and engaging handle **511** of payload **2150**. A lip **806** of payload retriever **800** extends through aperture **513** in handle **511** of payload **2150** to engage payload **2150**. FIG. **45**D is a perspective view of payload retriever **800** after removal of payload **2150** from pins **570** and **572** of payload holder **2060**.

(270) FIG. **46**A is a side view of payload retriever **800** attached to tether **2162** passing through channel passage **1433** of channel **1410** towards locking member **2039**. Locking member **2039** operates in a similar manner as locking member **2040** shown in FIGS. **45**A-D. Locking member **2039** extends over a latch on locking strap **2090** to lock locking strap **2090** into a locked position over handle **511** of payload **2150** and pins **570** and **572**. Locking strap **2090** operates in a similar manner as locking strap **2055** shown in FIGS. **45**A-D, except that locking strap **2090** includes holes **2010** and **2012** through which pins **570** and **572** extend. When positioned in the locked position shown in FIG. **46**A, locking strap **2090** serves to prevent the theft of payload **2150**, as well as prevent the opening of payload **2150** and any tampering with the contents within payload **2150**.

(271) FIG. **46**B is a perspective view of locking strap **2090** positioned over handle **511** and pins **570** and **572**; and FIG. **46**C is another perspective view of locking strap **2090** positioned over handle **511** and pins **570** and **572**. In FIGS. **46**B and **46**C, payload retriever **800** attached to tether **2162** has not yet passed over the movable end of locking member **2039** such that locking strap

2090 is still in a locked position over handle **511** of payload **2150** and pins **570** and **572**. FIG. **46**D is a perspective view of locking strap **2090** in an open position after payload retrieval **800** has passed over and depressed the movable end of locking member **2039** to unlatch locking strap **2090** and moved locking strap **2090** into an unlocked and open position shown in FIG. **46**D. The locking member **2039** may be spring-biased such that the movable end of locking member **2039** is biased towards the interior of channel **1410**. In addition, locking strap **2090** may be spring-biased into the open position shown in FIG. **46**D.

(272) FIG. **46**D is a perspective view of payload retriever **800** exiting the channel passage **1433** of channel **1410** and engaging handle **511** of payload **2150**. A lip **806** of payload retriever **800** extends through aperture **513** in handle **511** of payload **2150** to engage payload **2150** to effectuate payload retrieval by payload retriever **800** attached to tether **2162** and suspended from a UAV.

(273) Locking straps **2055** and **2090** described above are shown extending over handle **511** of payload **2150** and pins **570** and **572**. However, a locking strap or door could be configured to extend over not only handle **511** of payload **2150** and pins **570** and **572**, but also extend over the entire payload **2150**. A locking strap or door extending over the entire payload **2150** could be operated in the same manner as locking straps **2055** and **2090** to unlock a locking strap or door configured to extend over the entire payload **2150**.

(274) FIG. **47**A is a perspective view of combination payload retrieval and package pickup apparatus **2000**. However, combination payload retrieval apparatus **2000** could also be configured in the same manner as payload retrieval apparatus **1900** described above, as well as the other payload retrieval apparatuses described above. Payload retrieval apparatus **2000** includes tether engagers **2030** and **2032** which are used to engage a tether **2160** attached to a payload retriever **2160**, also referred to as payload retriever, shown in FIG. **47**B and described above. Tether engager **2030** includes member **2033** which provides mechanical support for tether engager **2030**, and provides other functions. In the same manner, tether engager **2032** includes member **2034** which provides mechanical support for tether engager **2032**, and provides other functions.

(275) Combination payload retrieval and package pickup apparatus **2000** also includes a package receptacle housing **2100** having package receptacles **2110** into which packages are received for subsequent package pickup, which can be driver pickup (by a driver in a vehicle) or curbside pickup (by a person picking up a package).

(276) Combination payload retrieval and pickup apparatus **2000** includes an autoloader assembly mounted to a base, which may be, but is not required to be, package receptacle housing **2100**. The autoloader assembly includes a payload holder **2060** configured to hold a payload **2150** for retrieval by a UAV, and also includes a channel coupled to the payload holder **2040** and configured to direct a payload retriever to the payload holder **2060**, the channel having a tether slot adapted for receiving a tether having a first end attached to the UAV and a second end attached to a payload retriever, wherein the channel is adapted to receive the payload retriever suspended from the UAV, and the tether slot in the channel allows for passage of the tether during movement of the payload retriever through the channel, wherein when the payload retriever exits the channel, the payload retriever is configured to engage a handle of the first payload and remove the first payload from the payload holder. The details of the channel are described in more detail above in the description of payload retrieval apparatus **1900** in FIGS. **42-44** and elsewhere.

(277) An extending member **2050** is positioned above the package receptacle housing **2100** and extends to payload holder **2060** which is adapted to secure a handle **2049** of payload **2150** where payload **2150** is in position for retrieval by a payload retriever suspended from a tether from a UAV. A payload receiver platform **2062** is also positioned on package receptacle housing **2100** that is configured to receive a payload delivered from a UAV. Combination payload retriever and package pickup apparatus **2000** shown in FIG. **47**A could also be provided without a package receptacle housing, such as payload retrieval apparatus **1900** described above. FIG. **47**B is a side view of combination payload retrieval and package pickup apparatus **2000** with payload **2150**

positioned on pins **570** and **572** on payload holder **2060**.
(278) FIG. **47**C illustrates payload holder **2060**, pins **570** and **572**, and payload **2150** are shown after having been retracted into a slot or drawer **2070** within the extending member **2050**. A door or sprung cover **2072** is shown after it has been closed over the slot or drawer **2070**. Door or sprung cover **2072** may be locked to prevent access to payload **2150** and its contents behind the door or sprung cover **2072**. FIG. **47**D is a side view of payload **2150** positioned within the slot or drawer **2070** behind locked door or sprung cover **2072**. Payload holder **2060** may be spring-loaded such that payload holder **2060** is biased into a forward position shown in FIGS. **47**A and **47**B. The payload holder **2060** may be latched within the extending member **2050** when the payload holder **2060** is in a retracted position as shown in FIGS. **47**C and **47**D. Further, door or sprung cover **2072** may be latched into the closed position shown in FIGS. **47**C and **47**D, and door or sprung cover **2072** may be spring-biased into the open position shown in FIGS. **47**A and **47**B.
(279) The payload holder **2060** and door or sprung cover **2072** may be unlatched using a locking member **2039** or **2040** when a payload retriever passes through the channel and depresses a movable end of the locking member **2039** or **2040** to allow for retrieval of payload **2150** by the payload retriever suspended from a tether from a UAV.
(280) In addition, instead of using a mechanical locking member **2039** or **2040** to unlatch locking straps **2055** and **2090**, and unlatch payload holder **2060** and door **2072**, a switch could be placed in the interior of channel **1960** which is tripped when the payload retriever passes over the switch and the switch sends a signal to unlatch the locking straps **2055** and **2090**, or to unlatch payload holder **2060** and door **2072**.
(281) Furthermore, a locking member **2039** or **2040**, or a switch within the interior of channel **1960** are not required to provide for the unlatching of locking straps **2055** or **2090**, or payload holder **2060** or door **2072**. Instead, the latches for those members could be unlatched using an electrical signal, WiFi, infrared, UAV proximity sensor or other sensor that is activated by a signal received from a tablet, phone, or computer. In this scenario, a servo actuated locking arm, locking coupler, cam latch, cam hook latch, hinge with an electrical switching function, etc. could also be used for the purpose of unlatching the locking straps **2055** or **2090**, or payload holder **2060** or door **2072**. In addition, each of these could also be unlatched manually. For example, if it is determined that the wrong payload is positioned on the payload holder, or an order is cancelled, the locking feature used to provide security for the payload may be unlocked manually to remove the payload from the payload holder. In this regard, a key could be used that extends into pivot **2042** of locking member **2040**, where the key is operable to move the movable end **2046** and first end **2044** of locking member **2040** to unlatch locking strap **2055** and unlock the payload **2150** for payload retrieval, as an example.
(282) FIG. **48**A is a perspective view of combination payload retrieval and package pickup apparatus **2000** with payload **2150** positioned on pins **570** and **572** on payload holder **2060**. The details of payload retrieval and package pickup **2000** is described above with respect to FIGS. **47**A-**47**D and therefore will not be further described here. FIG. **48**B is a side view of combination payload retrieval and package pickup apparatus **2000** with payload **2150** positioned on pins **570** and **572** on payload holder **2060**. Payload holder **2060** is rotatable into the autoloader assembly that includes extending member **2050**.
(283) FIG. **48**C is a perspective view of combination payload retrieval and package pickup apparatus **2000** after payload holder **2060** has been rotated 180 degrees such that the payload **2150** is positioned within extending member **2050** of the autoloader assembly. FIG. **48**D is a side view of combination payload retrieval and package pickup apparatus **2000** after payload holder **2060** has been rotated 180 degrees such that the payload **2150** is positioned within extending member **2050** of the autoloader assembly and latched into that position. After payload holder **2060** has been rotated 180 degrees such that the payload **2150** is positioned within extending member **2050** of the autoloader assembly, the payload **2150** is out of view and out of reach to prevent theft of payload

2150, or tampering with payload **2150** or its contents.

(284) The payload holder **2060** may be unlatched using a locking member **2039** or **2040** when a payload retriever passes through the channel and depresses a movable end of the locking member **2039** or **2040** to unlock and unlatch payload holder **2060** to rotate payload **2150** back into an exposed position as shown in FIGS. **48**A and **48**B to allow for retrieval of payload **2150** by the payload retriever suspended from a tether from a UAV.

(285) In addition, instead of using a mechanical locking member **2039** or **2040** to unlatch payload holder **2060** and rotate payload holder 180 degrees back into position for payload retrieval shown in FIGS. **48**A and **48**B, a switch could be placed in the interior of channel **1960** which is tripped when the payload retriever passes over the switch and the switch sends a signal to unlatch the payload holder **2060** and rotate payload **2150** back into an exposed position for payload retrieval.

(286) Furthermore, a locking member **2039** or **2040**, or a switch within the interior of channel **1960** are not required to provide for the unlatching of and rotation of payload holder **2060**. Instead, the latch for payload holder **2060** could be unlatched using an electrical signal, WiFi, infrared, UAV proximity sensor or other sensor that is activated by a signal received from a tablet, phone, or computer. In this scenario, a servo actuated locking arm, locking coupler, cam latch, cam hook latch, hinge with an electrical switching function, etc. could also be used for the purpose of unlatching and rotating payload holder **2060**.

(287) In addition, payload holder **2060** could also be unlatched manually. For example, if it is determined that the wrong payload is positioned on the payload holder, or an order is cancelled, the locking feature used to provide security for the payload may be unlocked manually to remove the payload from the payload holder. In this regard, a key could be used that extends into pivot **2042** of locking member **2040**, where the key is operable to move the movable end **2046** and first end **2044** of locking member **2040** to unlatch locking strap **2055** and unlock the payload **2150** for payload retrieval, as an example.

(288) FIG. **49**A is a perspective view of locking gate **2080** having bars **2082** positioned above payload **2150**; and FIG. **49**B is a side view of locking gate **2080** positioned above payload **2150**. Locking gate **2080** may also be positioned below payload **2150** where it may be raised into a position around the payload **2150** as shown in FIGS. **49**C and **49**D.

(289) FIG. **49**C is a perspective view of locking gate **2080** after being moved downwardly into a position around payload **2150** and latched into position; and FIG. **49**D is a side view of locking gate **2080** positioned around payload **2150**. Locking gate **2080** may be a two-dimensional gate (i.e., a flat gate) or may be three-dimensional gate, also referred to as a cage. After gate **2080** has been moved into position around payload **2150** as shown in FIGS. **49**C and **49**D, the payload **2150** is secured within gate **2080** and out of reach to prevent theft of payload **2150**, or tampering with payload **2150** or its contents.

(290) In the same manner as payload holder **2060** may be unlatched and rotated as described above with respect to FIGS. **48**A-D, gate **2080** may be unlatched using locking members **2039** or **2040**, or a switch within payload channel **1960**. Gate **2080** is spring-biased into a raised position. To release gate **2080** from a latched position around payload **2150** as shown in FIGS. **49**C and **49**D, gate **2080** may also be unlatched electronically or manually as described above regarding payload holder **2060**.

(291) FIGS. **50**A and **50**B illustrate another embodiment of a payload retrieval apparatus **5000** with a security feature to protect a payload **510** that is loaded on the payload retrieval apparatus **5000**. Similar to the other apparatuses, payload retrieval apparatus **5000** includes a payload holder **5030** and a channel **5050** configured to direct a payload retriever toward the payload. The payload retrieval apparatus also includes a pair of tether engagers **5002** configured to direct a tether toward an inlet end of the channel **5050**.

(292) The payload retrieval apparatus **5000** includes a payload bay **5090** that provides space for the payload **510** when it is mounted on the payload holder **5030**. To secure the payload **510** within the

payload bay **5090**, the payload retrieval apparatus **5000** includes a latchable door **5055** that may enclose the payload bay **5090** as shown in FIG. **50**B. By securing the entire payload **510** within the closed payload bay **5090**, this security feature prevents both unauthorized removal of the payload from the payload holder, but also destruction of the payload container to retrieve the contents inside. The closed payload bay **5090** also prevents loading of an unauthorized payload onto the payload holder **5030**, to prevent a UAV from accepting and delivering unauthorized objects.

(293) The door **5055** of the payload bay **5090** is held in place by a latch **5080** that is received in a catch **5056** on the door **5055**. The latch **5080** may be activated by a switch that is triggered as the payload retriever passes through the channel **5050**. Accordingly, when the payload retriever passes through the channel **5050**, the latch **5080** may be retracted and the door allowed to open. The payload **510** can then be removed from the payload holder **5030** as previously described.

(294) In some embodiments, the door **5055** may be mounted by a spring loaded hinge **5092** that is configured to move the door **5055** into an open position. This allows the door **5055** to automatically open when a payload retriever passes through the channel **5050** and activates the switch to retract the latch **5080**. With the door **5055** open the payload retriever can remove the payload **510** from the payload holder **5030** and be winched up to the UAV. Moreover, once the door **5055** is opened, a user will see that the payload retrieval apparatus **5000** is available for a new payload. The user can then load a new payload onto the payload holder **5030** and lift the door **5055** closed. Once the door **5055** is closed, the latch **5080** will hold it shut until the new payload is removed by another UAV.

(295) Having the door remain open once the latch **5080** is retracted may be particularly advantageous if the payload retrieval apparatus is part of a group of apparatuses, so that the user can be immediately informed of which of the apparatuses are available for a new payload and which are currently occupied. Alternatively, rather than having the door be spring-loaded open, the door **5055** may be configured to fall open under the force of gravity without the need for a spring.

(296) In other embodiments, the door **5055** may be spring loaded closed, so long as the UAV is able to remove the payload **510** from the payload bay **5090** when the door becomes unlatched, for example by tilting the door out of the way as the payload is winched upward on the tether. Biasing the door closed can help ensure that the door **5055** will be locked shut after a payload **510** is loaded onto the payload holder **5030**.

(297) The switch that operates to retract the latch **5080** may be configured similarly to the locking members described above. Alternatively, the switch may have another configuration. For example, FIGS. **51**A-**51**C illustrate a mechanism **5100** for retracting a latch **5180** that holds a door **5155** closed, though a similar mechanism could be used for unlocking other locking structures. The mechanism **5100** includes a switch **5120** that is configured as a protrusion that extends into the channel **5150**. The protrusion **5120** is coupled to the latch **5180** through an actuating linkage **5102** so that the depressing the protrusion **5120** causes the latch **5180** to retract, as shown in FIG. **51**B. In the illustrated embodiment, the actuating linkage **5102** includes a pivoting lever **5104** and a Bowden cable **5106**. As shown in FIG. **51**B, when the protrusion **5120** is pushed out of the channel **5150** by a payload retriever **800**, the protrusion **5120** rotates a lever which pulls the inner cable **5108** of the Bowden cable **5106**. The opposite end of the inner cable **5108** is secured to the latch **5180** and retracts the latch **5180**. As indicated by the two arrows, the door **5155** is then allowed to open.

(298) In some embodiments, the switch **5120** may be positioned within a cavity **5151** of the channel **5150**. This reduces the likelihood that the switch **5120** will be activated by objects other than the payload retriever **800**, as the object is unlikely to have the appropriate shape to activate the switch **5120**.

(299) For additional security, the door **5155** in FIG. **51**A is held in place by two latches **5180** and **5181**. Each latch **5180**, **5181** is independently operated by a respective protrusion **5120**, **5121**, and the second latch **5121** is coupled to the second protrusion **5121** by a separate actuating linkage

5101. Accordingly, the door **5155** is released only when both latches **5180**, **5181** are retracted. This provides added security, as a very particular shape is likely needed to depress both protrusions **5120**, **5121** at the same time. For example, FIG. **51**C shows an object **5111** that is small enough to fit within the cavity **5151** of the channel **5150** and is depressing the protrusion **5120** to retract the latch **5180**. However, the second protrusion **5121** is not depressed, and therefore the door **5155** remains shut. Of course, it is also possible to have the door held shut and released with a single activatable latch.
(300) While the switch that activates the locking system may physically extend into the channel, as shown in various embodiments described herein, in other embodiments, the switch may be positioned outside the channel but still be activated by a payload retriever passing through the channel. Such an embodiment is shown in FIGS. **52**A and **52**B. The mechanism **5200** of FIGS. **52**A and **52**B has a similar structure to mechanism **5100** of FIGS. **51**A and **51**B, and includes a spring loaded latch **5280** that holds a door **5255** shut. An actuating linkage **5202** including a Bowden cable **5206** pulls the latch **5280** to release the door **5255**. However, rather than using a lever that is activated by depressing a protrusion when the payload retriever passes through the channel, the inner cable **5208** of the Bowden cable **5206** is pulled by a magnetic element **5220** that forms an activatable switch. As the payload retriever **800** moves through the channel, a component on the payload retriever **800** causes the magnetic element **5220** to be pulled toward the channel **5250** and activate the linkage **5201**. The magnetic component on the payload retriever **800** may be another magnet or a ferromagnetic element. This configuration allows the switch to release a locked structure without placing any obstructions within the channel that may interfere with the passage of the payload retriever.
(301) Another embodiment that can provide security without any interference of the retrieval of the payload is shown in FIG. **53**. The payload retrieval apparatus **5300** shown in FIG. **53** includes a payload holder **5330** and a channel **5350** configured to direct a payload retriever toward the payload. The payload retrieval apparatus **5300** also includes a pair of tether engagers **5302** configured to direct a tether toward an inlet end of the channel **5350**.
(302) The payload retrieval apparatus **5300** includes a payload bay **5390** that is surrounded by a tall open-top enclosure **5370**. The opening **5372** in the top of the enclosure **5370** allows the payload **510** to be lifted out of the enclosure **5370** after it is secured on the payload retriever. However, the walls of the enclosure **5370** are sufficiently tall that a person cannot easily reach the payload and remove it. The amount of security provided by such a configuration may depend on the distance between the edge of the opening and the payload. For example, the walls may be taller, angled outward, or spaced further from the payload to hinder access to the payload. For example, the edge of the top opening of the payload bay may be spaced at least 24 inches, at least 30 inches, or at least 36 inches from the payload holder **5330**, e.g., from either pin of the payload holder. Additional security can also be provided by adding another wall between the tether engagers and the payload bay **5390**. While the opening at the top of the illustrated enclosure **5370** is unobstructed, in other embodiments, the payload bay may include a door that closes the opening at the top of the enclosure. For example, the payload bay may be provided with a door that swings upward when the payload is removed from the payload retrieval apparatus. For further security, the door may be configured to only swing outward and may include a flap that closes the opening when the door swings outward. This allows a payload to be removed by traveling between the door and the flap when the door swings upward, but prevents a person from reaching into the payload bay.
(303) To load a payload **510** on the payload holder **5330**, the payload bay **5390** may include a door **5392** for accessing the interior of the payload bay **5390**. The door **5392** may include a lock **5394** for limiting access to the payload holder **5330** and payload **510** only to authorized users.
(304) FIGS. **54**A and **54**B illustrate another embodiment of a locking element that secures the payload handle **511** of a payload **510** on a payload holder **5430**. The payload holder **5430** includes two pins **5432** configured to hold the payload handle **511** in a manner similar to those described

above. Once the payload handle **511** is on the pins **5432**, a locking bracket **5460** may be slid to cover the top of the handle **511** and close off the opening above the pins **5432**. Accordingly, the portion of the handle **511** that is above each pin **5432** is secured within a loop formed by the structure of the channel **5450**, the pin **5432** and the locking bracket **5460**.

(305) In order to release the payload **510**, the locking bracket **5460** may be activated to move to a release position, as shown in FIG. **54**B, when a payload retriever **800** passes through the channel **5450**. A portion of the locking bracket **5460** extends into the channel **5450** obstructing access through the channel **5450**. However, this portion of the locking bracket **5460** includes an angled surface **5462** that faces the upstream end of the channel **5450**. Accordingly, when the payload retriever **800** is drawn through the channel **5450**, as shown in FIG. **54**B, the payload retriever **800** engages the angled surface **5462** and pushes the locking bracket **5460** out of the channel **5450**, thereby allowing the payload retriever to pass through the channel. As the locking bracket **5460** is slid out of the channel **5450** it is also moved back from the pins **5432**, thereby clearing the opening above the pins **5432** and releasing the payload. As a result, the payload retriever **800** can pull the handle **511** off the pins **5432** and retrieve the payload **510**.

(306) FIGS. **55**A and **55**B illustrate another embodiment of a locking structure that includes a locking bracket **5560** that closes off the opening above the pins **5532** of the payload holder **5430**. In this embodiment, the locking bracket **5560** is deployed and moved into a locking position upon loading the payload **510** onto the payload holder **5530**. As shown in FIG. **55**A, the pins **5532** are configured to move between an upper unloaded position and a lower loaded position. Upon placing a payload handle **511** onto the pins **5532**, the pins **5532** can be pulled down into the loaded position and the weight of the payload **510** may hold the pins **5532** in the loaded position, as shown in FIG. **55**B.

(307) The moving pins **5532** are configured to cooperate with the locking bracket **5560** to move the locking bracket **5560** into the locked position when the payload **510** is loaded onto the pins **5532**. In particular, the locking bracket **5560** rotatable about a fulcrum **5562** and includes a lever **5564** that is driven by one of the pins **5532**. When the pin **5532** is moved to the lower position it pushes the lever **5564**, thereby rotating the locking bracket **5560** to contact the top of the pins and close off the opening that holds the payload handle **511**.

(308) While the illustrated embodiment of a locking structure that is actuated by loading the payload includes a bracket that closes the opening above the pins, other locking structures, such as covers, doors and boxes may similarly be actuated by loading the payload onto the payload holder. For example, an embodiment of a locking structure that includes a cover, such as that shown in FIGS. **45**A-**45**D, could be configured to swing closed upon loading the payload onto the payload holder.

(309) FIGS. **56**A-**56**C illustrate another embodiment of a locking structure **5600** that secures a payload onto a payload holder **5630** that is formed by a pair of pins **5632**. Components of the locking structure **5600** are shown in an exploded view in FIG. **56**A. The locking structure **5600** includes a locking bracket **5660** that closes next to the pins **5632** to prevent removal of a payload from the pins **5632**. The locking bracket **5660** includes a catch **5662** that is held closed by a latch **5680**. The latch **5680** is secured on a rotatable pin **5683** that is mounted on the channel structure **5652**. The latch **5680** is removable from the catch **5662** when the pin **5683** is rotated. To enable rotation, an actuating bumper **5664** is also attached to the rotatable pin **5683** and extends through an aperture **5654** into the interior of the channel **5650**. Accordingly, as a payload retriever passes through the channel **5650**, the actuating bumper **5664** is pushed out of the channel **5650**, which rotates the pin **5683**, thereby releasing the latch **5680** from the catch **5662**. Once the latch **5680** is released, the locking bracket **5660** can open, thereby releasing the payload from the pins **5632**.

(310) In order to open the locking bracket **5660**, the locking bracket **5660** may be rotatably secured on a hinge **5666** so that it can swing out of the way when the payload is to be removed. To encourage the locking bracket **5660** to swing open after the latch **5680** is released, the locking

bracket **5660** may be spring-loaded to an open position. In the illustrated embodiment, a torsion-spring **5668** is included in the hinge **5666** to urge the locking bracket **5660** open when it is not held by the latch **5680**. However, in other embodiments, other spring mechanisms may be used to urge the locking bracket **5660** open, such as a leaf spring or spring wire that is deformed when the locking bracket **5660** is closed.

(311) The rotatable pin **5683** is also spring-loaded toward a locked position, such that the force of the spring holds the latch **5680** in position to keep the locking bracket **5660** closed, as shown in FIG. **56**B. A cross-sectional side view of the locking mechanism **5600** is shown in FIG. **56**C. As shown, when a payload retriever **800** passes through the channel **5650**, it will push the actuating bumper **5664** out of the channel **5650** thereby rotating pin **5683** and removing latch **5680** from the catch **5662** on locking bracket **5660**. The torsion spring **5668** in hinge **5666** will then swing the locking bracket **5660** open, allowing a secured payload to be retrieved. As illustrated, the front side of the latch **5680** may be angled so that the locking bracket **5660** can be pushed into place until the latch **5680** passes into the catch **5662** and holds the locking bracket **5660** closed.

(312) FIGS. **57**A and **57**B illustrate another embodiment of a locking structure **5700** that secures a payload onto a payload holder that is formed by a pair of pins **5732**. The structure is similar to that of locking structure **5600** and includes a locking bracket **5760** that closes over a payload that is loaded on the pins **5732**. Again, the locking bracket **5760** may be rotatable on a hinge **5766** that is spring loaded open. The locking bracket **5760** also includes a catch **5762** that holds the locking bracket **5760** closed by cooperating with a rotating component that includes a latch **5780**. In contrast to the latch of structure **5600**, latch **5780** is part of a rotating component **5788** that is positioned on the side of the channel structure **5652**.

(313) Top, back, and side views of the rotating component **5788** are shown in FIG. **56**B. As shown, the latch **5780**, which is configured to be inserted into the catch **5762**, and includes a flat internal surface **5781**. The rotating component **5788** that forms the latch **5780** is spring-loaded closed so that the flat internal surface **5781** is held in engagement with the catch **5762** until the rotating component **5788** is pushed outward. The opposing outer surface **5782** of the latch **5780** is angled so that the locking bracket **5760** can push the latch **5780** against the spring force when the locking bracket **5760** is closed, until the latch **5780** moves into the catch **5762** and the rotatable component **5788** returns to its closed position.

(314) To enable rotation of component **5788** and release the locking bracket **5760**, the rotable component **5788** includes an actuating bumper **5764** that extends through an aperture **5754** in the side of the channel structure **5752**. In order to open the locking bracket **5760**, a payload retriever passing through the channel **5750** will push the actuating bumper **5764** out of the channel and thereby remove the latch **5780** from the catch **5742** of the locking bracket **5760**, which will in turn swing open.

(315) Various embodiments described herein use a payload retriever to unlatch a locking element. In such embodiments, the latches or locking elements may also be coupled to another mechanism to provide an alternative for releasing the locking elements, for example by using a physical or digital key.

(316) The particular arrangements shown in the Figures should not be viewed as limiting. It should be understood that other implementations may include more or less of each element shown in a given Figure. Further, some of the illustrated elements may be combined or omitted. Yet further, an exemplary implementation may include elements that are not illustrated in the Figures.

(317) Additionally, while various aspects and implementations have been disclosed herein, other aspects and implementations will be apparent to those skilled in the art. The various aspects and implementations disclosed herein are for purposes of illustration and are not intended to be limiting, with the true scope and spirit being indicated by the following claims. Other implementations may be utilized, and other changes may be made, without departing from the spirit or scope of the subject matter presented herein. It will be readily understood that the aspects of the

present disclosure, as generally described herein, and illustrated in the figures, can be arranged, substituted, combined, separated, and designed in a wide variety of different configurations, all of which are contemplated herein.

Claims

1. A payload retrieval apparatus comprising: a base; an autoloader assembly mounted to the base including: a payload holder configured to hold a payload for retrieval by an uncrewed aerial vehicle (UAV), a channel coupled to the payload holder and configured to direct a payload retriever suspended from the UAV to the payload holder, the channel having a tether slot adapted for receiving a tether having a first end attached to the UAV and a second end attached to the payload retriever, wherein the channel is adapted to receive the payload retriever, and the tether slot of the channel allows for passage of the tether during movement of the payload retriever through the channel; and a locking feature configured to lock access to the payload on the payload holder, the locking feature including a locking member that has a movable end, wherein the locking member has a first position in which the movable end extends through a wall of the channel into an interior of the channel for engagement by the payload retriever as it moves through the channel, wherein when the payload retriever contacts the movable end of the locking member, the movable end moves outwardly thereby moving the locking member to a second position and unlocking the payload on the payload holder; wherein the payload holder is positioned such that when the payload retriever exits the channel, the payload retriever engages a handle of the payload and removes the payload from the payload holder.
2. The payload retrieval apparatus of claim 1, wherein the movable end of the locking member is spring-biased into the interior of the channel.
3. The payload retrieval apparatus of claim 1, wherein the payload holder includes one or more pins configured to extend through a handle of the payload when the payload is positioned on the payload holder; wherein the locking features includes a locking strap configured to extend over the one or more pins and the handle of the payload; wherein the locking strap includes a latch which is latchable onto the autoloader assembly; and wherein the latch is unlatched when the payload retriever engages the movable end of the locking member and moves the locking member into the second unlocked position.
4. The payload retrieval apparatus of claim 3, where the locking strap also extends over the entire payload.
5. The payload retrieval apparatus of claim 1, wherein the payload holder includes one or more pins configured to extend through a handle of the payload when the payload is positioned on the payload holder; wherein the payload holder and payload are retractable into a slot in the autoloader assembly; wherein the locking features includes a sprung cover positioned over the payload holder and payload when the payload holder and payload have been retracted into the slot in the autoloader assembly; wherein the sprung cover includes a latch which is latchable onto the autoloader assembly; and wherein the latch of the sprung cover is unlatched when the payload retriever engages the movable end of the locking member and moves the locking member into the second unlocked position, and moves the payload holder and payload out of the slot and into position for payload retrieval.
6. The payload retrieval apparatus of claim 5, wherein the sprung cover is spring-loaded into an open position and the payload holder is spring-loaded into a position for payload retrieval by the payload retriever.
7. The payload retrieval apparatus of claim 1, wherein the payload holder includes one or more pins configured to extend through a handle of the payload when the payload is positioned on the payload holder; wherein the payload holder and payload are rotatable into a position within the extending member; wherein the locking feature includes a latch attached to the autoloader

assembly when the payload holder and payload have been rotated into position within the autoloader assembly; wherein the latch is unlatched when the payload retriever engages the movable end of the locking member and moves the locking member into the second unlocked position causing the payload holder and payload to rotate out of the autoloader assembly into position for payload retrieval.
8. The payload retrieval apparatus of claim 7, wherein the payload holder is spring-loaded to rotate into a position for payload retrieval by the payload retriever.
9. The payload retrieval apparatus of claim 1, wherein the payload holder includes one or more of pins configured to extend through a handle of the payload when the payload is positioned on the payload holder; wherein the locking feature includes a gate that extends over the payload holder and payload; wherein the gate includes a latch which is latchable onto the autoloader assembly; and wherein the latch of the gate is unlatched when the payload retriever engages the movable end of the gate and moves the locking member into the second unlocked position and moves the gate from extending over the payload holder and payload.
10. The payload retrieval apparatus of claim 9, wherein the gate is a three-dimensional cage.
11. The payload retrieval apparatus of claim 9, wherein the gate is spring-loaded into an open position where the gate does not extend over the payload holder and payload.
12. A method including: providing a payload retrieval apparatus having a base; an autoloader assembly mounted to the base including: a payload holder configured to hold a payload for retrieval by an uncrewed aerial vehicle (UAV), a channel coupled to the payload holder and configured to direct a payload retriever suspended from the UAV to the payload holder, the channel having a tether slot adapted for receiving a tether having a first end attached to the UAV and a second end attached to the payload retriever, wherein the channel is adapted to receive the payload retriever, and the tether slot of the channel allows for passage of the tether during movement of the payload retriever through the channel; and a locking feature configured to lock access to the payload on the payload holder, the locking feature including a locking member that has a movable end, wherein the locking member has a first position in which the movable end extends through a wall of the channel into an interior of the channel for engagement by the payload retriever as it moves through the channel, wherein when the payload retriever contacts the movable end of the locking member, the movable end moves outwardly thereby moving the locking member to a second position and unlocking the payload on the payload holder; and wherein the payload holder is positioned such that when the payload retriever exits the channel, the payload retriever engages a handle of the payload and removes the payload from the payload holder; and causing the payload retriever to pass through the channel and engage the movable end of the locking member to unlock access to the payload holder and payload.
13. A payload retrieval apparatus comprising: a base; an autoloader assembly mounted to the base including: a payload holder configured to hold a payload for retrieval by an uncrewed aerial vehicle (UAV), a channel coupled to the payload holder and configured to direct a payload retriever suspended from the UAV to the payload holder, the channel having a tether slot adapted for receiving a tether having a first end attached to the UAV and a second end attached to the payload retriever, wherein the channel is adapted to receive the payload retriever, and the tether slot of the channel allows for passage of the tether during movement of the payload retriever through the channel; and a locking feature configured to lock access to the payload on the payload holder; and wherein the payload holder is positioned such that when the payload retriever exits the channel, the payload retriever engages a handle of the payload and removes the payload from the payload holder.
14. The payload retrieval apparatus of claim 13, wherein the payload holder includes one or more pins configured to extend through a handle of the payload when the payload is positioned on the payload holder; wherein the locking feature includes a locking strap which extends over the one or more pins and the handle of the payload; wherein the locking strap includes a latch which is

latchable onto the autoloader assembly; and wherein the latch is unlatchable to unlock access to the payload holder and payload.

15. The payload retrieval apparatus of claim 13, wherein the payload holder includes one or more pins configured to extend through a handle of the payload when the payload is positioned on the payload holder; wherein the payload holder and payload are retractable into a slot in the autoloader assembly; wherein the locking feature includes a sprung cover positioned over the payload holder and payload when the payload holder and payload have been retracted into the slot in the autoloader assembly; wherein the sprung cover includes a latch which is latchable onto the autoloader assembly; and wherein the latch of the sprung cover is unlatchable to unlock access to the payload holder and payload and move the payload holder and payload out of the slot and into position for payload retrieval.

16. The payload retrieval apparatus of claim 15, wherein the sprung cover is spring-loaded into an open position and the payload holder is spring-loaded into a position for payload retrieval by the payload retriever.

17. The payload retrieval apparatus of claim 13, wherein the payload holder includes one or more pins configured to extend through a handle of the payload when the payload is positioned on the payload holder; wherein the payload holder and payload are rotatable into a position within the autoloader assembly; wherein the locking feature includes a latch attached to the autoloader assembly when the payload holder and payload have been rotated into position within the autoloader assembly; wherein the latch is unlatchable to unlock access to the payload holder and payload and rotate the payload holder and payload out of the autoloader assembly into position for payload retrieval.

18. The payload retrieval apparatus of claim 17, wherein the payload holder is spring-loaded to rotate into a position for payload retrieval by the payload retriever.

19. The payload retrieval apparatus of claim 13, wherein the payload holder includes one or more of pins configured to extend through a handle of the payload when the payload is positioned on the payload holder; wherein the locking feature includes a gate that extends over the payload holder and payload; wherein the gate includes a latch which is latchable onto the autoloader assembly; and wherein the latch of the gate is unlatchable to unlock access to the payload holder and payload and move the gate from extending over the payload holder and payload.

20. The payload retrieval apparatus of claim 19, wherein the gate is spring-loaded into an open position where the gate does not extend over the payload holder and payload.

21. The payload retrieval apparatus of claim 19, wherein the gate is a three-dimensional cage.

22. The payload retrieval apparatus of claim 13, wherein the locking feature includes a locking member that is manually unlockable.

23. The payload retrieval apparatus of claim 22, wherein a key is manually operable to move the locking member into an unlocked position.

24. The payload retrieval apparatus of claim 13, wherein the locking feature includes a locking member that is electronically unlockable.

25. A method including: providing a base; an autoloader assembly mounted to the base including: a payload holder configured to hold a payload for retrieval by an uncrewed aerial vehicle (UAV), a channel coupled to the payload holder and configured to direct a payload retriever suspended from the UAV to the payload holder, the channel having a tether slot adapted for receiving a tether having a first end attached to the UAV and a second end attached to the payload retriever, wherein the channel is adapted to receive the payload retriever, and the tether slot of the channel allows for passage of the tether during movement of the payload retriever through the channel; and a locking feature configured to lock access to the payload on the payload holder; and wherein the payload holder is positioned such that when the payload retriever exits the channel, the payload retriever engages a handle of the payload and removes the payload from the payload holder; and unlocking the locking feature to provide access to the payload holder and payload for payload retrieval by the

payload retriever.

26. A payload retrieval apparatus comprising: a base; an autoloader assembly mounted to the base including: a payload holder configured to hold a payload for retrieval by an uncrewed aerial vehicle (UAV), a channel coupled to the payload holder and configured to direct a payload retriever suspended from the UAV to the payload holder, the channel having a tether slot adapted for receiving a tether having a first end attached to the UAV and a second end attached to the payload retriever, wherein the channel is adapted to receive the payload retriever, and the tether slot of the channel allows for passage of the tether during movement of the payload retriever through the channel; and a locking feature configured to lock access to the payload on the payload holder, the locking feature including a switch disposed along the channel and configured to be activated by the payload retriever as it moves through the channel, wherein when the payload retriever activates the switch, the locking member unlocks the payload on the payload holder; and wherein the payload holder is positioned such that when the payload retriever exits the channel, the payload retriever engages a handle of the payload and removes the payload from the payload holder.